Precise Localisation for Vineyard Pruning Operations

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ABSTRACT

This thesis is bloody brilliant.

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Thanks mum and dad!

PREFACE

A preface is useful for chatting to the reader about why you wrote this thesis, what is original about your work, and what material you have published.

NOTATION

If you have a lot of mathematics it is a good idea to have a notation section and a glossary of the symbols employed in the thesis. In this section you might also include definitions and acronyms. The LaTeX package nomencl is useful for constructing a glossary.

Chapter 1

INTRODUCTION

1.1 RESEARCH CONTEXT

This thesis is situated within the broader research initiatives of the University of Canterbury Computer Vision Research Group (UCVision), specifically contributing to 'The Occlusion Project'. This project utilises autonomous field robots to capture multi-view imagery of vineyards and orchards, which are subsequently processed *offline* to generate high-fidelity 3D reconstructions using 3D Gaussian Splatting (3DGS). These models serve as the "digital twin" required for complex agricultural tasks such as precision yield estimation and automated labour.

A primary objective of this project is the automation of vineyard pruning. In this workflow, pruning decisions and robotic arm path planning are computed within the high-fidelity 3DGS environment. Unlike conventional pruning systems that rely on lower-resolution, real-time sensor data for immediate modelling and execution, the UCVision workflow decouples data collection from intervention. While this allows for superior structural analysis of the vine, it introduces a critical spatial synchronisation problem: the realignment of global and local coordinate frames.

For the robot to execute a pre-computed pruning plan, it must bridge the gap between its real-time physical state and the coordinate system of the offline model. This necessitates a robust SLAM framework capable of solving three distinct sub-problems:

1. **Global Relocalisation:** Navigating the platform back to the specific plant or row identified in the modelling phase.

2. **Strategic Positioning:** Aligning the mobile base within the physical workspace constraints of the robotic arm to ensure all required cutpoints are reachable.

3. **Sim-to-Real Registration:** Achieving millimeter-level alignment between the robot's onboard perception and the 3D model. This ensures that the simulated path-planner's coordinates map precisely to the physical vine structures in real life.

This thesis addresses this capability gap by developing a SLAM and relocalisation framework that facilitates this alignment. By synchronizing the robot's perceived position with both the physical environment and the pre-existing 3D model, this research enables the transition from static digital twins to active, autonomous robotic intervention in the field.

1.2 RESEARCH MOTIVATION

This research is primarily motivated by the need to enable autonomous agricultural robots to perform complex manipulation tasks, specifically robotic pruning. Unlike simple navigation, pruning requires a high-fidelity localisation and mapping (SLAM) framework that can provide the millimetre-level precision necessary for a robotic arm to interact with thin, overlapping vine structures. Developing a system that can maintain this accuracy in the presence of dynamic foliage and repetitive, unstructured row patterns is a critical bottleneck in automating viticulture and horticulture.

The demand for such technology is underscored by the economic landscape of New Zealand's primary industries. As of 2026, horticulture export revenue is forecast to reach a record $9.2 billion, with wine production constituting $2.3 billion [Industries 2025]. However, the sector faces chronic labour shortages and rising operational costs, with manual pruning alone accounting for around 23% of annual vineyard expenses [2025 Marlborough Vineyard Monitoring Report]. The 2025 Vineyard Monitoring Report published by the Ministry for Primary Industries indicates that during the 2024/2025 season, expenditure on manual pruning increased, while spending on machine pruning declined by 21%. This shift is attributed to grower concerns regarding vine damage caused by current mechanical stripping machines [2025 Marlborough Vineyard Monitoring Report]. These concerns highlight a critical limitation of current commercial machine-based pruning systems: insufficient precision to meet vineyard managers' operational requirements. Overcoming these precision challenges would enable robotic labour to provide a scalable response to ongoing workforce constraints, thereby supporting the long-term sustainability of New Zealand's high-value horticultural crops.

Beyond the immediate application to pruning, this work seeks to advance the research objectives of the UCVision Research Lab. By improving SLAM robustness in vegetated environments, this thesis contributes to the lab's broader mission of "seeing through occlusion" and tracking plant growth across multiple stages. The frameworks developed for semi-structured orchard conditions extend directly to other challenging environments such as forestry or search-and-rescue, providing a foundation for autonomous systems in visually complex terrain.

1.3 RESEARCH OBJECTIVES

The main objective of this research is to develop a precise localisation system for autonomous vineyard robots that enables accurate spatial alignment between the robot, the 3D models, and the environment.

Sub-objectives include:

- Quantify the localisation accuracy required for robotic pruning.
- Design scientific experiments and establish quantitative performance metrics to evaluate localisation accuracy.
- Achieve the necessary localisation accuracy from sub-objective 1 in vineyard environments.

- Demonstrate the practical application of the developed system by enabling precise robotic manipulation tasks such as pruning within the vineyard environment.
- Validate localisation performance under varying vineyard conditions including GNSS dropout, lighting changes, and seasonal variations.

1.4 THESIS STRUCTURE

This thesis is structured as follows: Chapter 2 establishes the theoretical and practical foundations of robot localisation for vineyard manipulation tasks. Chapter 3 presents the development and evaluation of a multi-sensor SLAM system for initial mapping and coarse localisation within vineyard environments. Chapter 4 describes the precise-alignment methodology, wherein robot pose estimates are refined through registration with pre-existing 3D models using stereo depth imaging, dense LiDAR, and Iterative Closest Point (ICP) algorithms. Chapter 5 concludes with a synthesis of key findings and recommendations for future research.

Chapter 2

BACKGROUND

2.1 ROBOT LOCALISATION

2.1.1 Introduction

Robot localisation is the process of estimating a robot's pose — position and orientation — within a given reference frame using sensory information. It is a prerequisite for all navigation and manipulation tasks that require interaction with the physical environment, and is tightly coupled with mapping; together they constitute Simultaneous Localisation and Mapping (SLAM) [Ullah et al 2024, Leonard and Durrant-Whyte 1991].

This section traces the historical foundations of robot localisation, structured around three persistent challenges that have shaped the field and that remain directly relevant to the methods presented in this thesis. The first is the accumulation of pose error under dead reckoning, which motivates a probabilistic treatment of uncertainty. The second is the need to represent and fuse uncertain, heterogeneous sensor measurements into a coherent spatial model. The third is the tension between short-range alignment and global map consistency. The foundational contributions reviewed here — probabilistic uncertainty modelling [Smith and Cheeseman 1987], occupancy-based sensor fusion [Elfes 1989], the Iterative Closest Point (ICP) algorithm [Besl and McKay 1992], and pose graph optimisation [Lu and Milios 1997] — constitute the algorithmic backbone of the SLAM system developed in chapter 3 and the precise-alignment pipeline described in chapter 4.

2.1.2 The Limits of Deterministic Localisation

The first mobile robots capable of autonomous reasoning operated under deterministic models: each commanded action was assumed to execute perfectly, and the robot's pose was updated accordingly. Shakey the Robot (1966–1972) localised through discrete semantic labels and open-loop dead reckoning, with landmark correction that depended on the environment conforming exactly to a pre-modelled structure [Nilsson 1984]. Moravec's Stanford Cart extended this to geometric localisation via stereo vision, maintaining explicit 3D feature positions but associating no uncertainty with them; errors accumulated silently, requiring approximately five hours for a 20-metre indoor traverse [Moravec 1980, Moravec 1990].

These systems illustrate the core failure mode of deterministic localisation: uncorrected motion errors compound over time, and landmark-based correction is unreliable whenever the observed environment departs from assumed models. In semi-structured outdoor environments such as vineyards — where plant geometry changes seasonally, features repeat along rows, and no guaranteed prior map exists — this fragility is not a boundary case but the normal operating condition. The response was to replace deterministic pose estimates with probabilistic representations that could explicitly track and propagate uncertainty.

2.1.3 Uncertainty Representation and Sensor Fusion

The shift to probabilistic localisation was formalised by Smith and Cheeseman [Smith and Cheeseman 1987], who proposed representing each spatial relationship as an *approximate transformation*: a pose estimate paired with a covariance matrix encoding second-order uncertainty. Their framework introduces two fundamental operations. *Compounding* propagates uncertainty through sequential transformations: when two uncertain poses are composed, the resulting covariance grows as a function of both input uncertainties, formalising what Shakey and the Cart demonstrated empirically — that drift accumulates with each successive motion. *Merging* reduces uncertainty by fusing independent estimates of the same quantity via a minimum-variance weighted combination, yielding a posterior covariance smaller than either input. In a subsequent paper, Smith, Self, and Cheeseman extended this framework to the Stochastic Map, in which robot and landmark poses are jointly maintained in a single state vector — the formulation underpinning Extended Kalman Filter (EKF) SLAM.

Building on this probabilistic foundation, Elfes introduced occupancy grids [Elfes 1989]: a spatial representation in which each cell of a tessellated lattice maintains a probabilistic occupancy estimate, updated incrementally from raw sensor data through an explicit sensor model. A key insight is that individual sensor modalities carry complementary uncertainty profiles — sonar provides low range uncertainty but poor angular resolution, while stereo cameras offer good angular resolution but depth uncertainty that grows with distance. Representing both in a common occupancy-grid format allows them to be fused into a map with lower aggregate uncertainty than either source alone, as illustrated in fig. 2.1. Elfes also noted that fusing sensor views into a coherent global map requires accurate motion estimation, since dead-reckoning uncertainty compounds in precisely the manner formalised by Smith and Cheeseman. This co-dependence between localisation accuracy and map quality is the defining characteristic of SLAM, and directly motivates the choice in chapter 3 to fuse LiDAR odometry with an inertial measurement unit (IMU) whose complementary error characteristics reduce drift under the same merging principle.

2.1.4 Point Cloud Registration

Relative pose estimation between sensor frames requires a method for aligning observations acquired at successive robot positions. Besl and McKay's Iterative Closest Point (ICP) algorithm [Besl and McKay 1992] frames registration as an optimisation problem: starting from an initial alignment, it alternates between assigning each source point to its nearest neighbour

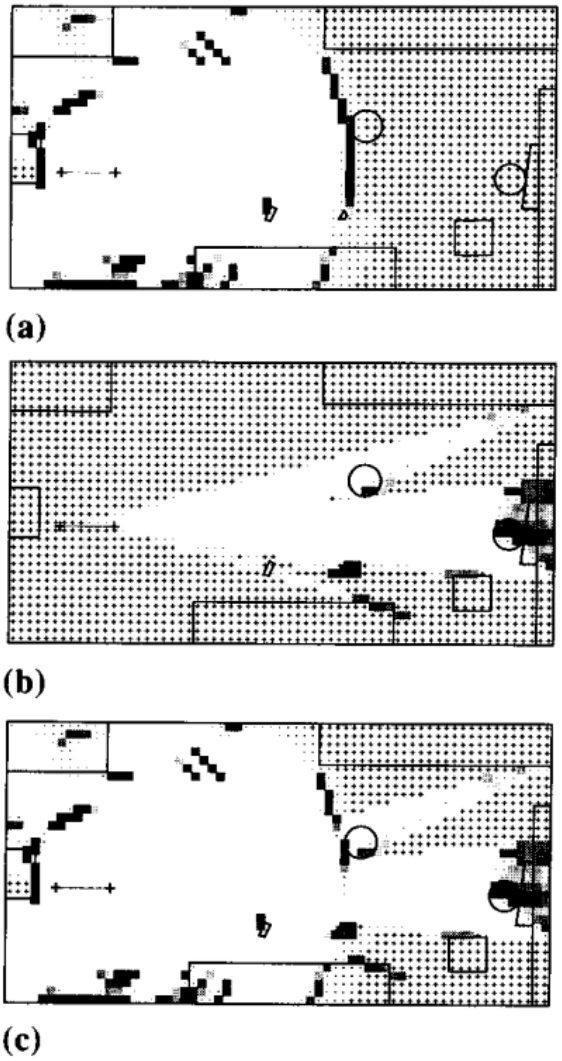


Figure 2.1: Sensor integration of sonar and scanline stereo. Occupancy grids generated separately for sonar (a) and scanline stereo (b), as well as the integrated map (c) obtained through sensor fusion, are shown. Occupied regions are marked by shaded squares, empty areas by dots fading to white space, and unknown spaces by + marks [Elfes 1989].

in the target cloud and solving for the rigid transformation that minimises the mean squared distance over all correspondences. Convergence to a local minimum is guaranteed; however, the solution is sensitive to initial conditions and requires sufficient overlap between the two clouds. Subsequent variants — notably point-to-plane error metrics [Chen and Medioni 1991] and the probabilistic Generalised-ICP formulation [Segal et al 2009] — improved convergence speed and robustness in environments lacking strong planar structure.

ICP is the direct registration engine used in the precise-alignment pipeline described in chapter 4, where a real-time point cloud of a physical vine is aligned against the SfM-derived reference geometry extracted from a 3D Gaussian Splatting model. The sensitivity of ICP to initialisation and partial overlap is a central practical constraint addressed in that chapter.

2.1.5 Pose Graph Optimisation and Global Consistency

Probabilistic uncertainty modelling and local scan matching together support accurate short-range pose estimation, but do not address a deeper limitation: errors accumulated along a long trajectory cannot be corrected retroactively by filtering methods. EKF-based SLAM integrates past measurements into a global state and then discards those measurements; over extended trajectories, linearisation errors accumulate and the filter becomes internally inconsistent, with different regions of the map updated independently and unable to be reconciled.

Lu and Milios addressed this by introducing pose graph optimisation [Lu and Milios 1997]. In a pose graph, each estimated robot pose is a node, and each relative spatial constraint — derived from scan matching or odometry — is an edge. The optimisation objective is to find the configuration of all nodes that best satisfies the full set of edge constraints simultaneously. Because the entire history of pose estimates is retained, a single loop-closure detection event can propagate corrections backward through the trajectory, recovering a globally consistent map. The fundamental contrast with filtering is that past states are not discarded: global consistency is achieved through a batch optimisation rather than a recursive update, at the cost of a more demanding solve.

For long traversals through vineyard rows, where cumulative drift along the row axis would otherwise render multi-row maps geometrically inconsistent, this distinction is operative. The SLAM system presented in chapter 3 employs pose graph optimisation as its back-end, enabling loop-closure corrections to enforce global consistency across traversals of multiple vineyard rows.

2.1.6 Summary

The history reviewed in this section identifies three challenges that persist in the vineyard localisation problem addressed by this thesis. First, pose error accumulates under dead reckoning and must be explicitly modelled and propagated — demanding a probabilistic treatment of estimation. Second, no single sensor is sufficient for robust localisation; each modality carries characteristic errors, and heterogeneous sensor fusion governed by the merging operations formalised by Smith and Cheeseman is necessary to reduce aggregate uncertainty. Third, local consistency from scan matching and global consistency from loop-closure correction are distinct requirements, and satisfying both simultaneously requires a graph-based optimisation back-end.

The systems developed in chapter 3 and chapter 4 address these three challenges in sequence. The following section examines how these challenges manifest within vineyard and orchard environments, where domain-specific constraints have motivated a distinct body of applied SLAM research.

2.2 ROBOT LOCALISATION IN VINEYARD AND ORCHARD ROBOTICS

Unlike structured industrial settings, agricultural environments are "semi-structured" environments characterised by homogeneous structures, seasonal variability, and dynamic occlusion [Hroob et al 2021]. Moreover, the reliance on Global Navigation Satellite Systems (GNSS) often falls short in orchards as dense canopies can attenuate satellite signals or induce multipath errors, making standard GNSS insufficient for centimetre-level precision tasks [Tang et al 2024]. Therefore, research has focused on developing technologies that use semantic understanding, geometric constraints, and multi-sensor fusion to operate reliably in these GNSS-denied environments [Tang et al 2024].

2.2.1 Environmental Constraints and Challenges

Semi-structured agricultural environments create unique constraints that pose a challenge for generic SLAM algorithms designed for structured urban environments.

Perceptual Aliasing and Geometric Repetitiveness Commercial crops in vineyards and orchards are planted in long, parallel rows with uniform spacing to maximise yield and facilitate machine access. The long uniform rows create a “corridor effect”, especially for LiDAR-based systems. The corridor effect is characterised by the absence of distinct geometric features along the longitudinal position of the robot (the direction of movement down the row), such that the longitudinal axis becomes unobservable, which can lead to significant drift in the estimated trajectory as the robot cannot determine if it has travelled one metre forward or five metres forward [Khan et al 2025]. Visual systems face a similar challenge wherein standard feature descriptors (e.g., ORB, SIFT) extracted from one part of a row are statistically indistinguishable from those in another, resulting in erroneous feature matching and false-positive loop closure [Papadimitriou et al 2022].

Dynamic and Unstructured Environment Many SLAM systems make an assumption of a static world, however orchards and vineyards are dynamic environments. Short-term dynamics such as wind-induced canopy motion, workers, and machines can pose challenges to mapping and loop closure. Long-term dynamics include seasonal changes, wherein a map constructed during the dormant winter season can differ greatly compared with the full-canopy summer season. This can pose challenges to long-term localisation and map maintenance [Schmidt et al 2024].

Furthermore, the uneven terrain can cause camera shake and wheel slippage that introduces high-frequency noise into odometry estimates [Zhang et al 2026].

2.2.2 LiDAR SLAM

LiDAR SLAM is often favoured due to its illumination invariance, but standard scan-matching algorithms like ICP often fail in symmetric vineyard rows due to the aforementioned perceptual aliasing along the longitudinal axis.

VineSLAM tries to address the perceptual aliasing problem by grouping point clouds into specific vineyard features: semi-planes (for the continuous canopy/trellis) and points (for trunks/posts). It utilises a Particle Filter that fuses these features; semi-planes offer lateral constraints to keep the robot centred, and point features provide longitudinal constraints along the row. Comparative studies show VineSLAM outperforms general-purpose algorithms like LeGO-LOAM in symmetric corridors [Aguiar et al 2022].

Tree-SLAM detects individual tree trunks within the point clouds, mapping the environment as a sparse graph of tree landmarks rather than a dense point cloud. This method achieved localisation errors as low as 18 cm for individual trees (less than 20% of standard planting distance) in apple and pear orchards [Rapado-Rincon and Kootstra 2025]. This object-level approach improves navigation but also facilitates agronomic tasks such as per-tree yield monitoring [Rapado-Rincon and Kootstra 2025].

The April-LIO-SAM framework tightly couples LiDAR-Inertial Odometry (LIO-SAM) with AprilTags acting as artificial landmarks. April tags provide absolute pose constraints that constrain the drift from LiDAR odometry. Experimental validation in orchards demonstrated that with tags spaced at 3 to 5 metres, the system achieves a root-mean-square error (RMSE) of approximately 0.057 m, maintaining stability even under occlusion and varying illumination [Guan et al 2025]. Another multi-sensor system achieved positioning errors of 0.57 cm in pear orchards, using visual data for loop closure and LiDAR for geometric structure [Tang et al 2024].

Benchmarking Datasets Progress in the field is accelerated by open-source datasets tailored to agricultural challenges:

- **MAgro Dataset:** Collected in apple and pear orchards using LiDAR, stereo cameras, and RTK-GPS. It is designed to benchmark loop closure and localisation under varying weather and illumination conditions [Schmidt et al 2024] [Marzoa Tanco et al 2024].
- **Bacchus Long-Term (BLT) Dataset:** A unique dataset capturing a vineyard over a 7-month period. It supports research into "lifelong SLAM," challenging algorithms to handle the drastic morphological changes of the canopy from winter dormancy to summer harvest [Polvara et al 2024].
- **AgriLiRa4D:** A recent multi-modal dataset (LiDAR, Camera, 4D Radar) collected via UAV over diverse terrains (flat, hilly), facilitating research into radar-based fusion and aerial coverage paths [Zhan et al 2026].

2.2.3 Summary

The development of SLAM for orchards and vineyards has evolved from the application of generic algorithms to the design of domain-specific architectures. Current research prioritises semantic SLAM to filter dynamic agricultural noise and object-level mapping (e.g., Tree-SLAM) to resolve geometric ambiguity. Furthermore, the integration of multi-sensor fusion (e.g., April-LIO-SAM) has proven essential for achieving the centimetre-level accuracy required for precision tasks. Future work is increasingly driven by long-term autonomy, leveraging datasets like BLT to enable robots that can adapt to the seasonal morphological changes of the canopy [Tang et al 2024, Polvara et al 2024].

Within vineyard robotics, pruning represents the most demanding localisation sub-problem. In addition to the perceptual aliasing and sensor-fusion challenges described above, it imposes explicit millimetre-level accuracy requirements on pose estimation and introduces constraints particular to the dormant season, reviewed in section 2.3.

2.3 ROBOT LOCALISATION IN VINEYARD PRUNING

Pruning typically occurs in the dormant season (winter), which introduces a unique robot localisation challenge in the form of geometric sparsity and the lack of distinctive visual features caused by the absence of foliage [Aguiar et al 2022]. The environment consists of thin, bare branches that provide a low density of LiDAR points and few distinctive visual textures for cameras to localise on [Ahmed et al 2023]. Furthermore, without leaves to block the view, sensors often pick up distant "noise" from the sky or neighbouring rows, leading to errors in depth estimation and foreground segmentation [You et al 2022]. The following analysis details the strategies used for spatial alignment with vines, specifically in the context of robotic pruning platforms.

2.3.1 Spatial Alignment Modalities for Pruning Vines

While traditional passive vision remains the baseline for high-resolution 3D modelling, it faces significant geometric and environmental hurdles. To address the ill-conditioned configurations of binocular systems – specifically when linear canes align with the stereo baseline – Botterill et al. employ a trinocular rig consisting of three 1.3-megapixel cameras arranged in a triangular configuration [Botterill et al 2017]. This allows for the dynamic selection of the mathematically optimal camera pair for specific cane segments, a level of geometric robustness that passive binocular systems lack.

However, even optimised trinocular systems remain vulnerable to outdoor lighting variability. This necessitates a shift toward active sensing, either through illumination or direct range-finding. Systems like "Bumblebee" integrate active light stereo cameras to generate dual point clouds, which are then refined via Iterative Closest Point (ICP) algorithms to maintain invariance against ambient shadows [Navone et al 2025, Silwal et al 2021]. In contrast, Felicetti et al. bypass the lighting and correspondence problem entirely by utilising downward-facing 3D LiDAR [Felicetti et al 2025]. By applying a minimalist depth-thresholding algorithm within a

specific range (D_{min} to D_{max}, typically 250–500 mm), this approach sacrifices the fine-grained topological "expert system" understanding provided by Botterill's trinocular rig in favour of raw speed, processing a vine in just 7.3 seconds [Felicetti et al 2025, Botterill et al 2017].

The computational backend of these systems further distinguishes their operational philosophies. Botterill et al. prioritise spatial stability for robotic manipulation through a heavy incremental bundle adjustment pipeline. By utilising the *g2o* graph optimisation framework and robust Pseudo-Huber cost functions to penalise large residuals, they achieve trajectory errors below 1%, though at a higher computational cost. Recent paradigms, however, suggest that such global optimisation may be unnecessary for effective pruning. Navone et al. and the "Archie Jnr" platform utilise deep panoptic segmentation models (e.g., YOLOv8-seg or Detectron2) to extract semantic coordinates directly in image space [Navone et al 2025, Hizatate et al 2025].

By registering these masks with depth data, these neural architectures achieve an mAP_{50-95} of 87.2% for pruning point detection (a candidate-identification metric rather than a measure of 3D positioning error), effectively bridging the gap between the high-fidelity reconstruction of Botterill et al. and the high-speed obstacle-navigation of Felicetti et al. [Botterill et al 2017, Felicetti et al 2025, Navone et al 2025]. Ultimately, while the trinocular approach provides the most detailed 3D map for precise spur pruning, the integration of active illumination and semantic segmentation offers a more computationally viable path toward commercial field speeds.

Table 2.1: Comparison of Perception Modalities for Vineyard Localisation

System	**Modality**	**Spatial Accuracy**	**Speed/Throughput**
Botterill et al. [Botterill et al 2017]	Trinocular Stereo	High (Millimetre)	120 s/vine
Silwal et al. [Silwal et al 2021]	Active Stereo	High (Millimetre)	137 s/vine
Felicetti et al. [Felicetti et al 2025]	3D LiDAR	Moderate (Depth-filter)	7.3 s/vine
Archie Jnr [Navone et al 2025]	RGB-D (YOLOv8)	High (3D Coordinates)	15.2 s/cut point

2.3.2 Summary

Existing robotic pruning platforms demonstrate that millimetre-level spatial alignment is achievable, but they share a common architectural assumption: perception, planning, and execution are tightly coupled within a single real-time pipeline. The UCVision approach departs from this assumption by decoupling the data-collection phase from the intervention phase, generating a high-fidelity 3D Gaussian Splatting model offline before any manipulation is attempted. This decoupling introduces a spatial synchronisation problem that no existing pruning platform directly addresses: given a pre-computed pruning plan expressed in the coordinate frame of an offline 3D model, how can a robot executing that plan achieve the sub-centimetre registration accuracy necessary to transfer planned cut-points to physical vine structures? This gap motivates the contributions of chapter 3 and chapter 4.

Chapter 3

SLAM

3.1 INTRODUCTION

This chapter focuses on the development and implementation of a SLAM system designed to operate independently of frequent GNSS updates. The primary objective is to ensure high-fidelity localisation in GNSS-denied or GNSS-degraded scenarios. Furthermore, a critical requirement for this system is temporal robustness; that is, the SLAM front-end must remain invariant to fluctuating visual appearances, enabling continuous 24/7 operation across both diurnal and nocturnal cycles, different weather conditions, and plant growth cycles. Thirdly, as aforementioned, vineyard and orchard rows form long, symmetric, corridor-like structures which limits the geometric diversity available to scan-matching algorithms, making them susceptible to ambiguity and longitudinal drift. The system presented in this chapter is therefore not intended as a general-purpose SLAM solution; it is a specialised architecture targeted at the structural and perceptual characteristics of row-crop environments such as vineyards and orchards.

3.1.1 Motivation

The necessity for this research stems from the operational limitations of the current UCVision Rover localisation system. Prior iterations of this platform utilised an Extended Kalman Filter (EKF) heavily reliant on Real-Time Kinematic (RTK) GPS data. While precise under open skies, this dependency proved non-viable for robust field operations where tree canopies, topography, and public RTK base station blackouts frequently cause signal multipath or dropout. Adverse weather further compounds this unreliability: heavy precipitation increases tropospheric delay errors [Solheim et al 1999] and multipath reflections from wet vegetation [Nievinski and Larson 2014], while ionospheric disturbances — which are more frequent during solar weather events and magnetic storms — introduce rapid carrier-phase cycle slips that break the integer ambiguity fix and degrade the RTK solution from centimetre-level fixed accuracy to decimetre-level float accuracy or worse [Wielgosz et al 2005]. In practice, a significant proportion of field scans conducted with the UCVision Rover were rendered unusable due to the inability to acquire or sustain a consistent RTK fix, causing the localisation system to fail entirely.

Transitioning to a LiDAR-centric SLAM approach, tailored to these agricultural environments, provides three distinct advantages:

- **GNSS-independent localisation:** By shifting the primary motion estimation to scan-matching and loop closure, the system becomes less reliant on RTK.
- **Corridor robustness:** The architecture presented here specifically addresses the structural challenges of row-crop environments through vineyard-focused parameterisation and sensor fusion suited to row-structured terrain.
- **Downstream utility:** Beyond localisation, the high-density LiDAR maps generated can be used for additional tasks such as landmark detection (e.g., vine trunks and trellis posts), spatial alignment of Gaussian splats captured from opposing sides of a row, and global registration of data across multiple vineyard bays.

3.2 RELATED WORK

The development of Simultaneous Localisation and Mapping (SLAM) has shifted from laboratory settings to complex, real-world deployments across autonomous driving, aerial vehicles, and subterranean exploration. Due to the limitations of external positioning systems such as GNSS in dynamic or urban environments, Light Detection and Ranging (LiDAR) has become the dominant sensing modality, providing high-fidelity, active 3D geometry independent of ambient illumination. The state of the art in this field is defined by LiDAR-Inertial Odometry (LIO), which tightly couples LiDAR measurements with high-frequency inertial data from an Inertial Measurement Unit (IMU) to compensate for motion distortion and maintain observability in geometrically degenerate scenes [Koide et al 2024]. Modern LIO research is primarily distinguished by a bifurcation of techniques into two dominant optimisation philosophies: Error-State Kalman Filtering (ESKF) and Factor Graph Optimisation (FGO). This review traces the architectural evolution of LIO, comparing and contrasting leading algorithms such as LIO-SAM, FAST-LIO2, GLIM, and Super-LIO to synthesise the current trends in computational efficiency and global consistency.

3.2.1 Modern LiDAR-SLAM Architectures

The foundation of modern LiDAR SLAM rests on the seminal LOAM (LiDAR Odometry and Mapping) framework [Zhang and Singh 2014], which introduced the critical concept of extracting and registering geometric features, specifically classifying points into “edge” and “planar” surfaces based on curvature. LOAM’s split-process architecture separated high-frequency odometry from low-frequency mapping. Subsequent work, such as LeGO-LOAM [Shan and Englot 2018], refined this by optimising ground points separately. However, these systems were loosely-coupled and feature-dependent, leading to inevitable drift over long trajectories and fragility in unstructured environments (e.g., foliage or feature-poor corridors).

The transition to tight-coupling was facilitated by the advancement of IMU Pre-integration theory [Forster et al 2017]. This re-parameterisation allows IMU integration to be expressed as a relative motion constraint (e.g., $\Delta\mathbf{R}_{ij}$) that is independent of the initial velocity and pose estimate. This innovation decouples the high IMU rate from the optimisation rate, making

it computationally feasible for graph-based methods to efficiently optimise state biases and velocities over large time windows.

The result of this evolution is LIO-SAM [Shan et al 2020], which established the feature-based factor graph benchmark for global consistency. LIO-SAM builds a factor graph incorporating LiDAR odometry, IMU pre-integration, and explicit loop closure factors. To manage computational complexity, it employs a "sliding window" local map for scan-to-map matching. Despite its accuracy and robustness compared to predecessors, its reliance on LOAM-style feature extraction and a standard KD-tree for neighbor searching limits its performance in geometrically degenerate environments.

The current state of the art is defined by two architectural tracks: those prioritising extreme computational efficiency via filtering, and those prioritising global consistency and robustness via dense graph optimisation.

Filter-Based Direct Odometry Algorithms in the embedded track are designed to minimize CPU usage, making them suitable for low-power ARM processors on platforms like small drones. These systems rely on the Error-State Kalman Filter (ESKF) and are primarily odometry engines that estimate state and covariance, but are susceptible to drift over time as past states are marginalized.

FAST-LIO2 [Xu et al 2022] is a widely adopted benchmark for computational efficiency, eschewing feature extraction in favour of registering raw points directly. Its two principal innovations are the Incremental KD-Tree (iKD-Tree), which supports dynamic map updates with amortised $O(\log N)$ complexity, and a Kalman Gain Inversion that reduces the filter update cost from the measurement dimension to the 18-dimensional state vector, enabling kHz-rate processing on modest hardware. Point-LIO [He et al 2023] extends this paradigm by updating the EKF state for every individual point as it arrives from the sensor, yielding a processing bandwidth bounded only by the sensor's sampling rate. This point-wise update strategy enables tracking under aggressive motion where conventional scan-accumulation methods fail due to intra-frame distortion. Super-LIO [Wang et al 2025b] represents the most recent advance in this track, targeting the map data structure rather than the estimator itself. It introduces the **OctVox** representation — a hybrid combining a spatial hash map with internal octrees — to enforce strict point density constraints, and pairs this with a Heuristic-Guided k-Nearest Neighbour (HKNN) search that further reduces per-scan processing time. Reported benchmarks indicate lower CPU utilisation than FAST-LIO2, suggesting OctVox-based representations may define the next generation of resource-constrained LIO.

Graph-Based Global Consistency Systems in this track prioritize long-term mapping quality and robustness by leveraging high-performance hardware, particularly embedded GPUs.

GLIM (Global LiDAR Inertial Mapping) [Koide et al 2024] represents a distinct departure from CPU-centric design, instead leveraging GPU parallelism to achieve dense, globally consistent mapping. Scan matching in GLIM is implemented via GPU-accelerated Voxelised Generalised Iterative Closest Point (VGICP) factors, which allow the system to incorporate nearly all points per frame and thereby retain geometric constraints that feature-based methods routinely

discard. A further key contribution is the Submap Endpoint Mechanism, in which IMU pre-integration factors are inserted as stiff constraints on unobservable degrees of freedom within each submap. This structural guarantee preserves registration quality even when the LiDAR geometry is degenerate — for example in featureless tunnels or elevator shafts. For back-end estimation, GLIM employs a fixed-lag smoother that continuously optimises a bounded window of recent poses, combining the accuracy benefits of iterative re-linearisation with bounded memory consumption, in contrast to full-history SLAM approaches.

This trade-off between computational efficiency and global consistency is summarized in the comparison table below. The most significant recent innovations have occurred in the map data structure domain (iKD-Tree, OctVox) for the filter-based track, and in hardware-software co-design (GLIM's GPU acceleration) for the graph-based track.

Table 3.1: Comparative Analysis of State-of-the-Art LIO Systems

Algorithm	**Backend**	**Map Structure**	**Loop Closure**	**Robustness (Degeneracy)**	**Efficiency (CPU)**
LIO-SAM	Factor Graph	Static KD-Tree	Yes	Moderate	Low
FAST-LIO2	Iterated ESKF	iKD-Tree	No	High (Direct)	High
Point-LIO	Iterated ESKF	iKD-Tree	No	Very High	Moderate
DLIO	Geometric Obs.	KD-Tree/iVox	No	High	High
GLIM	Factor Graph	GPU Voxelmap	Yes	**Extreme**	Low (GPU)
Super-LIO	Iterated ESKF	**OctVox**	No	High	**Very High**

Super-LIO is effective for projects constrained to low-power CPUs that can tolerate bounded drift, while GLIM excels at high-fidelity mapping applications where long-term global consistency and robustness to severe degeneracy are paramount, provided a GPU accelerator is available.

3.2.2 SLAM Hyperparameter Optimisation

The optimisation of SLAM systems has evolved from manual tuning into rigorous hyperparameter optimisation (HPO) and statistical space exploration. As robots transition from the lab into industrial, agricultural and extraterrestrial domains, stability, robustness and generalisation become pre-eminent [Wei et al 2026]. Many modern SLAM systems are configured by a complex set of interdependent parameters that influence behaviour from high-level feature extraction to back-end pose graph optimisation [Han et al 2020]. The challenge of configuring a high-dimensional space, often referred to as the curse of dimensionality, poses a significant hurdle in tuning these SLAM systems. The first challenge is finding a subset of points from a high-dimensional search space that accurately represents the underlying performance landscape [Han et al 2020].

Search Space Sampling One such method for sampling high-dimensional search spaces is Latin Hypercube Sampling (LHS). LHS, as formally introduced by Michael McKay in 1979, was

used as a variance reduction technique to address the inefficiencies of Monte Carlo simulation (MCS) in high-dimensional domains [Chrisman 2014]. In MCS, sample points are generated independently, which often leads to "clusters" and "voids" in disparate regions, which can neglect critical narrow regions of optimal performance [Deutsch and Deutsch 2012]. LHS addresses this by enforcing a stratification of the marginal distributions of each input variable.

LHS divides the range of each of the d variables into n equiprobable intervals. In a standard uniform distribution of $[0, 1]$, these strata are defined as $A_i = \left(\frac{i-1}{n}, \frac{i}{n}\right]$ for $i = 1, \ldots, n$ [Chrisman 2014]. To generate a LHS sample, one value is drawn from each interval for each dimension. These values are then combined into vectors such that each n sample point is the only representative in each axis-aligned hyperplane. A comparison of LHS, MCS and Grid Search is detailed in Table 3.2.

Sampling Attribute	**Monte Carlo (MCS)**	**Grid Search**	**Latin Hypercube (LHS)**
Variable Coverage	Random; susceptible to clustering	Uniform but sparse for high d	Guaranteed univariate uniformity [Chrisman 2014]
Sample Efficiency	Low; requires large n for stability	Very low; n^d scaling	High; independent of d [Deutsch and Deutsch 2012]
Bias/Variance	Unbiased; high variance	Biased to grid points	Consistent; asymptotically unbiased [Packham and Schmidt 2008]
Search Structure	Unstructured	Rigidly structured	Near-random with low discrepancy [Chrisman 2014]

Table 3.2: Comparison of Sampling Strategies

The advantage of LHS in SLAM parameterisation comes from its capacity to represent the parameter space with fewer samples than traditional methods. In a SLAM system with ten parameters, a grid search with only three levels per parameter would require $3^{10} = 59,049$ evaluations. Testing each sample on a real trajectory quickly becomes time prohibitive. LHS can explore the same search space with fewer points while maintaining that each parameter is tested at different levels [Deutsch and Deutsch 2012]. However, while uniformity is guaranteed across each axis, it does not guarantee that the interior of the hypercube is filled uniformly. As a result, researchers have developed Latin Hypercube Sampling with Multidimensional Uniformity (LHSMDU) to address the limitations of univariate stratification [Deutsch and Deutsch 2012]. LHSMDU employs a sequential elimination process, where a large pool of MCS candidates are generated, and those that are too close in the multidimensional space are pruned [Deutsch and Deutsch 2012]. This method maximises the minimum inter-point distance, maximising the spread of samples throughout the multi-dimensional space.

Often the SLAM parameters have physical or co-dependent constraints, such as turn radii or computational resource limits. Standard LHS assumes independent variables, but extensions

like Latin Hypercube Sampling with Dependence (LHSD) allow for the incorporation of copula-based dependence structures [Packham and Schmidt 2008]. This method enforces constraints on the sampling strategy so that the search does not enter undefined/invalid regions of the search space.

Robustness, Generalisation and Overfitting Often, it is not difficult to find a high-performance SLAM configuration for a single dataset. The real challenge is finding a configuration that generalises across a diverse set of environments. This is the common problem in machine learning of overfitting to a dataset, and is a widespread bottleneck to reliable and robust deployment of robotic systems [Soares et al 2025].

Different SLAM architectures are more sensitive to overfitting to specific environmental characteristics. For example, visual SLAM systems are more sensitive to overfitting to different lighting conditions, and LiDAR SLAM systems are more sensitive to overfitting to specific geometric characteristics. For example, a LiDAR SLAM parameter set tuned for a feature-rich, static hallway will likely adopt aggressive feature matching thresholds and high confidence in loop closure. When transitioned to an open field with sparse landmarks and faster movement, these same parameters may cause the system to lose tracking or generate false-positive loop closures [soumyadip 2024]. Some general examples of environmental factors, SLAM failure modes and impacts on tuning are illustrated in Table 3.3. This highlights the importance of avoiding overfitting to a single dataset, as characteristics such as "low light" for VSLAM can skew the optimisation process in a direction that is unsuitable for other lighting conditions.

Environmental Factor	**VSLAM Failure Mode**	**LiDAR SLAM Failure Mode**	**Impact on Tuning**
Low Light	Feature extraction failure [10]	Minimal (Active sensing)	Requires adaptive thresholds [10]
Dynamic Objects	Tracking drift; ghosting [Soares et al 2025]	Segmentation errors [soumyadip 2024]	Necessitates robust loss kernels [Soares et al 2025]
Low Texture	Tracking loss [19]	Geometric degeneracy [soumyadip 2024]	High dependency on sensor fusion [Wei et al 2026]
High Speed	Motion blur [Xu et al 2024]	Point cloud skew [soumyadip 2024]	Requires higher frame rates [Xu et al 2024]

Table 3.3: Impact of Environmental Factors on SLAM Performance

Traditional optimisation focuses on minimising a scalar error metric, such as Absolute Trajectory Error (ATE), on a single "representative" dataset – typically one of the prevailing SLAM benchmark datasets such as KITTI or TUM RGB-D. However, this approach tends to "chase" the noise of that specific trajectory [Kandasamy et al], resulting in parameters that are essentially "hard-coded" to that specific sequence of sensor observations and noise realisations of the training data.

3.3 PROPOSED METHODS

The following subsections describe the methods developed and integrated to achieve globally consistent, ENU-referenced localisation in vineyard environments. These contributions build on the GLIM framework surveyed in the preceding section and address the key challenges of reference frame alignment, GNSS outlier rejection, and parameter optimisation.

To begin this investigation, we implemented GLIM [Koide et al 2024]. We chose GLIM as it had infrastructure in place to add extension modules to incorporate new sensors and capabilities. GLIM provides a global callback slot mechanism that allows access to the internal states of the mapping process and insertion of additional constraints into the factor graph. GLIM also eliminated sensor-specific processes, meaning it can be applied to a wide variety of range sensors such as spinning-type LiDAR, non-repetitive scan LiDAR, solid-state LiDAR and even RGB-D cameras.

GLIM state estimation is created in the local odometry frame. For our use in the wider research project, we required state estimation in the local ENU reference frame, which is a surveyed-in point at each of our farms. Thus, we needed a procedure to align the GLIM reference frame with the existing local ENU reference frame.

Firstly, we time synchronise the on-board GPS unit with the LiDAR and with the system clock. This allows us to associate corresponding GNSS points with the submaps or keyframes of LiDAR frames that GLIM registers. We aggregate a set of corresponding GNSS coordinates with keyframe coordinates, and then implement the Kabsch algorithm [Kabsch 1976] to find the relative transform between the two point sets by minimising the root-mean-square deviation between the two point sets.

The Kabsch algorithm relies on the least-squares method to minimise the root-mean-square deviation (RMSD) between two point sets. Since the objective function squares the distances between paired points, outliers produce disproportionately large influences on the final transformation matrix. Thus, for global alignment, it is important that outliers are filtered out before executing the Kabsch algorithm.

3.3.1 Geometric Consistency-Based Outlier Rejection

To remove spurious GNSS measurements prior to initialisation, we apply a geometric consistency check that exploits the rigidity of the GNSS–IMU transformation. The method assumes that, for inlier measurements, pairwise distances between GNSS observations should be preserved when mapped into the local map frame via the estimated poses.

Let $\{\mathbf{g}_i\}_{i=1}^N$ denote GNSS positions expressed in the ENU frame, and $\{\hat{\mathbf{g}}_i\}_{i=1}^N$ denote the corresponding GNSS positions expressed in the map frame, obtained by transforming IMU poses with a fixed extrinsic calibration. For each point i, we evaluate its consistency with all other points $j \neq i$ by comparing pairwise Euclidean distances in both frames as shown in Equation (3.1):

$$|\|\mathbf{g}_i - \mathbf{g}_j\| - \|\hat{\mathbf{g}}_i - \hat{\mathbf{g}}_j\|| < \tau, \tag{3.1}$$

where τ is a distance consistency threshold reflecting accepted SLAM error and GNSS noise.

This approach was chosen over stochastic methods like RANSAC [Fischler and Bolles 1981] or iterative M-estimators because it takes advantage of the invariance of pairwise Euclidean distances under rigid body transformations, which means that the method remains entirely deterministic and independent of any initial pose estimate or probabilistic modelling.

To formalise the inlier classification, we define a pairwise consistency indicator c_{ij} in Equation (3.2):

$$c_{ij} = \begin{cases} 1 & \text{if } |\|\mathbf{g}_i - \mathbf{g}_j\| - \|\hat{\mathbf{g}}_i - \hat{\mathbf{g}}_j\|| < \tau \\ 0 & \text{otherwise} \end{cases} \tag{3.2}$$

A measurement is classified as an inlier if the fraction of other points with which it satisfies the above constraint exceeds a minimum inlier ratio ρ, satisfying the condition:

$$\frac{1}{N-1} \sum_{j \neq i}^{N} c_{ij} > \rho. \tag{3.3}$$

GNSS multipath errors [Wen et al 2018] can often create erroneous clusters of points that are mutually consistent, and therefore are not filtered out by conventional statistical outlier methods. However, the majority-based criterion in Equation (3.3) ensures that points are only kept if they belong to the global rigid consensus, effectively filtering out clusters of mutually consistent but globally erroneous points. After classification, only inlier GNSS measurements and their corresponding map-frame estimates are preserved for subsequent processing.

3.3.2 Planar Geometry Quality Assessment in 2D

Subsequently to outlier rejection, we assess the geometric quality of the remaining point set under the assumption of planar motion by analysing its distribution in the horizontal plane. This analysis is motivated by the sensitivity of the Kabsch algorithm to degenerate point configurations: near-collinear or poorly distributed trajectories produce ill-conditioned cross-covariance matrices, leading to unstable or biased transformations. By detecting such configurations before alignment, we ensure the SVD is well-conditioned and that the estimated transformation is numerically reliable. In particular, we enforce minimum requirements on translational baseline ($\geq 5.0\,\text{m}$), planar spread, rotational excitation, and numerical conditioning.

Given a set of 3D points $\{\mathbf{p}_i\}_{i=1}^{N}$, only the horizontal components are considered by projecting each point onto the xy-plane, yielding $\mathbf{q}_i = [x_i, y_i]^\top$. If fewer than three points are available, the geometry is deemed insufficient and the assessment terminates early.

The projected points are assembled into a data matrix $\mathbf{M}$ as defined in Equation (3.4):

$$\mathbf{M} = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_N \end{bmatrix} \in \mathbb{R}^{2\times N}, \tag{3.4}$$

which is centred by subtracting the mean $\bar{\mathbf{q}}$ from each column. Singular value decomposition is then applied to the centred matrix as shown in Equation (3.5):

$$\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \tag{3.5}$$

yielding two singular values $\sigma_1 \geq \sigma_2$ that characterise the principal spatial extents of the trajectory in the plane.

The condition number κ is defined in Equation (3.6):

$$\kappa = \frac{\sigma_1}{\sigma_2}, \tag{3.6}$$

and is used as a degeneracy metric, with large values indicating near-collinear motion. The geometry is considered non-degenerate if $\kappa < 20.0$.

To evaluate spatial coverage, a 2D spread ratio r is computed according to Equation (3.7):

$$r = \frac{\sigma_2}{\sigma_1}, \tag{3.7}$$

which measures how well the trajectory excites both planar directions. A minimum spread ratio of $r \geq 0.15$ is required. Additionally, a minimum displacement constraint is enforced by requiring both singular values to exceed a baseline threshold of 5.0 m. The spatial distribution is considered sufficient only if both the spread ratio and displacement conditions are satisfied.

Rotational excitation is evaluated independently by computing the angular span of the point set in the horizontal plane. The geometry is considered to provide sufficient rotational constraint if the angular span exceeds 45.0°.

The final geometry quality assessment combines these criteria, reporting whether the point set exhibits sufficient planar spread, adequate rotational excitation, and non-degenerate conditioning. This 2D-based evaluation provides a robust and numerically stable measure of geometric suitability for planar motion scenarios.

3.3.3 GLIM Reference Frame Alignment with Local ENU

Given two corresponding point sets $\{\mathbf{a}_i\}_{i=1}^N, \{\mathbf{b}_i\}_{i=1}^N \subset \mathbb{R}^3$, the goal is to estimate a rigid-body transformation $(\mathbf{R}, \mathbf{t}) \in \mathrm{SO}(3) \times \mathbb{R}^3$ that minimises the sum of squared alignment errors as defined in the following objective function:

$$\min_{\mathbf{R},\mathbf{t}} \sum_{i=1}^{N} \|\mathbf{b}_i - (\mathbf{R}\mathbf{a}_i + \mathbf{t})\|^2, \quad \text{s.t. } \det(\mathbf{R}) = 1. \tag{3.8}$$

First, the centroids of both point sets are computed as:

$$\boldsymbol{\mu}_A = \frac{1}{N}\sum_{i=1}^{N} \mathbf{a}_i, \qquad \boldsymbol{\mu}_B = \frac{1}{N}\sum_{i=1}^{N} \mathbf{b}_i. \tag{3.9}$$

The point sets are then centred by subtracting their respective centroids:

$$\tilde{\mathbf{a}}_i = \mathbf{a}_i - \boldsymbol{\mu}_A, \qquad \tilde{\mathbf{b}}_i = \mathbf{b}_i - \boldsymbol{\mu}_B. \tag{3.10}$$

A 3×3 cross-covariance matrix $\mathbf{H}$ is formed as:

$$\mathbf{H} = \sum_{i=1}^{N} \tilde{\mathbf{b}}_i \tilde{\mathbf{a}}_i^\top. \tag{3.11}$$

The singular value decomposition (SVD) of $\mathbf{H}$ is computed in Equation (3.12):

$$\mathbf{H} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top. \tag{3.12}$$

To ensure a proper rotation and avoid reflections, a diagonal correction matrix $\mathbf{D}$ is introduced:

$$\mathbf{D} = \mathrm{diag}(1,\, 1,\, \mathrm{sign}(\det(\mathbf{U}\mathbf{V}^\top))). \tag{3.13}$$

The optimal rotation $\mathbf{R}$ is then given by:

$$\mathbf{R} = \mathbf{U}\mathbf{D}\mathbf{V}^\top. \tag{3.14}$$

Finally, the translation $\mathbf{t}$ is recovered according to Equation (3.15):

$$\mathbf{t} = \boldsymbol{\mu}_B - \mathbf{R}\boldsymbol{\mu}_A. \tag{3.15}$$

The resulting rigid transformation $(\mathbf{R}, \mathbf{t})$ aligns the source point set to the target point set in a least-squares sense. This static transformation is published to the ROS2 transform tree [Macenski et al 2022] to link the local ENU frame to the GLIM mapping frame.

3.3.4 GNSS Prior Factor Construction

This procedure augments an existing SLAM factor graph [Grisetti et al 2010] with GNSS-based prior constraints that relate submap poses to global position measurements. Each GNSS measurement is incorporated as a probabilistic constraint on the estimated submap pose through a known sensor extrinsic calibration.

Let $\mathbf{T}_{\text{map}}^{\text{enu}} \in SE(3)$ denote the rigid transformation from the local ENU frame to the GLIM map frame, and $\mathbf{p}_{\text{imu}}^{\text{gps}} \in \mathbb{R}^3$ the fixed translation from the IMU frame to the GNSS antenna frame. For each correspondence between a submap and a GNSS measurement, the following steps are performed.

First, the GNSS position $\mathbf{p}_{\text{enu}} \in \mathbb{R}^3$ is extracted in the ENU frame and transformed into the map frame as shown in Equation (3.16):

$$\mathbf{p}_{\text{map}} = \mathbf{T}_{\text{map}}^{\text{enu}}\, \mathbf{p}_{\text{enu}}. \tag{3.16}$$

Next, a diagonal Gaussian noise model is constructed from the reported GNSS covariance. Let $\boldsymbol{\sigma} \in \mathbb{R}^3$ denote the standard deviations obtained from the square root of the covariance diagonal. These values are scaled by a configurable factor α and converted to a covariance

matrix in Equation (3.17):

$$\mathbf{\Sigma}_{\text{gps}} = \text{diag}\Big((\alpha\boldsymbol{\sigma})^2\Big)\,. \tag{3.17}$$

Given a submap pose $\mathbf{T}^{\text{sub}}_{\text{map}} \in SE(3)$, the predicted GNSS antenna position in the map frame is expressed in Equation (3.18):

$$\hat{\mathbf{p}}_{\text{map}} = \mathbf{T}^{\text{sub}}_{\text{map}}\,\mathbf{p}^{\text{gps}}_{\text{imu}}. \tag{3.18}$$

This formulation is valid as the GLIM submap origin is defined at the IMU center, such that the extrinsic offset $\mathbf{p}^{\text{gps}}_{\text{imu}}$ represents the antenna position relative to the submap origin. An expression-based factor is then constructed to minimize the residual between the predicted and measured positions under the noise model $\mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_{\text{gps}})$:

$$\mathbf{r}(\mathbf{T}^{\text{sub}}_{\text{map}}) = \|\hat{\mathbf{p}}_{\text{map}} - \mathbf{p}_{\text{map}}\|^2_{\mathbf{\Sigma}_{\text{gps}}}\,. \tag{3.19}$$

This factor directly constrains the corresponding submap pose variable in the factor graph, contributing to global consistency and LiDAR-inertial odometry drift correction. In effect, this factor constrains the GLIM-predicted position of the GPS antenna using the measured GNSS position at that time.

3.3.5 Robust SLAM Parameterisation via Iterative Region Refinement

The optimisation of Simultaneous Localisation and Mapping (SLAM) parameters is a high-dimensional, non-convex problem where the objective is to identify a parameter vector Θ that minimises localisation error across heterogeneous environments [Han et al 2020]. This subsection proposes a model-free, heuristic region-refinement algorithm that transitions from global stochastic exploration to localised deterministic refinement. The strategy assumes that optimal configurations cluster within specific sub-regions of the hyper-dimensional search space Ω, which are iteratively identified and contracted using statistical distribution analysis.

3.3.5.1 Problem Formulation

We define the SLAM system as a complex function $f(X, \theta)$, where X represents raw sensor data (e.g., LiDAR scans and inertial measurements) and Θ is the d-dimensional parameter vector. In a typical LiDAR-inertial setup, Θ includes parameters such as LiDAR resolution and registration strategy. The objective is to find Θ^* such that:

$$\Theta^* = \arg\min_{\Theta\in\Omega} \mathcal{J}(\Theta) \tag{3.20}$$

where $\mathcal{J}$ is a robust objective function representing the consensus performance across K distinct motion profiles.

3.3.5.2 Sampling via Latin Hypercube Strategy

To explore the initial parameter manifold, we utilise Latin Hypercube Sampling (LHS) as an alternative to traditional grid search or independent Monte Carlo (MC) sampling [Deutsch and Deutsch 2012]. LHS is technically categorised as a stratified sampling design [Chrisman 2014]. For a sample size N in d dimensions, the range of each parameter θ_j is partitioned into N equiprobable intervals for $j = 1, \ldots, d$ [Chrisman 2014]. The mathematical constraint — ensuring exactly one sample per row and column of the grid — minimises the "clustering" phenomenon common in pure random sampling [Deutsch and Deutsch 2012]. This stratification ensures that the marginal distribution of each parameter is sampled uniformly, achieving a lower variance than independent Monte Carlo sampling for the same sample size [Packham and Schmidt 2008].

Implementation and Normalisation Parameter vectors are first normalised to the unit hypercube $[0, 1]^d$ [Deutsch and Deutsch 2012]. To ensure reproducibility, the pseudo-random permutations used to construct the hypercube are seeded deterministically. The samples are subsequently scaled to physical ranges using:

$$\theta_{j,i} = \theta_{j,\min} + u_{j,i}(\theta_{j,\max} - \theta_{j,\min}) \tag{3.21}$$

where $\theta_{j,\min}$ and $\theta_{j,\max}$ represent the physical lower and upper bounds for parameter j.

3.3.5.3 The Multi-Dataset Consensus Framework

A primary challenge in SLAM optimisation is the risk of "over-specialisation", where parameters become overfitted to the unique noise characteristics of a single trajectory [Soares et al 2025]. We implement a Consensus Survival Framework to prioritise reliability across datasets.

In some situations, standard "Mean-Performance Optimisation" can hide brittle configurations that perform well on average but fail catastrophically on a single sequence [Kandasamy et al]. However, in this setting, catastrophic failure typically results in excessively high amplitude errors, which effectively inflate the mean normalised error. That being said, some particular parameter configurations can lead to excessive GPU draw, resulting in a runtime CUDA out-of-memory error. If the parameter set does not complete processing the trajectory on any dataset k, it is assigned an infinite cost and discarded from the survivor pool. This ensures the survival of only those configurations that meet the computational requirements of the system.

Formal Objective Function For the "surviving" configurations, the objective function $J(\theta)$ is defined as the mean normalized error across the valid sequences (3.22):

$$J(\theta) = \begin{cases} \frac{1}{K}\sum_{k=1}^{K} E_k(\theta) & \text{if success} \in \{D_1, \ldots, D_K\} \\ \infty & \text{otherwise} \end{cases} \tag{3.22}$$

where $E_k(\theta)$ is the primary performance metric for dataset k.

3.3.5.4 Iterative Region Refinement (IRR)

The optimisation proceeds through an "Iterative Contraction" or "Funnel" strategy. This mirrors the reduction of a search radius in traditional Trust Region Methods [Nocedal and Wright 2006], concentrating the sampling density into the most promising basins of attraction.

Statistical Search Space Contraction At the conclusion of each phase i, the top-performing configurations are used to calculate new search boundaries for phase $i+1$. We employ the 10th and 90th percentiles of the elite set as robust estimators for the update rule in Equation (3.23):

$$\Omega_{j,i+1} = [P_{10}(\theta_{j,\text{top}}), P_{90}(\theta_{j,\text{top}})]. \tag{3.23}$$

This quantile-based reduction is superior to using absolute Min/Max values because it is resilient to "lucky" outliers — configurations that may perform well due to noise interactions rather than parameter stability.

Refinement Phase Schedule The optimisation is structured into four phases, increasing the selection pressure (lower survival rate) as the search space contracts, as detailed in table 3.4:

Table 3.4: Evolutionary Phases of the Sampling Algorithm

Phase	Objective	Iterations (N)	Survival Rate	Description
Phase 1	Thinning	100	30%	Broad exploration identifying stable regions.
Phase 2	Narrowing	100	20%	Convergence on local minima.
Phase 3	Refinement	100	10%	High-density sampling within basins.
Phase 4	Fine-tuning	75	1.3%	Final tuning to find top candidate.

This incremental strategy is distinct from Successive Halving (SHA) [Jamieson and Talwalkar 2015], as it preserves a continuous distribution of potential solutions rather than discarding them in discrete early-stopping steps, which is critical for SLAM where early performance does not always correlate with final drift.

3.3.5.5 Optimisation Metrics

For the optimisation of the base GLIM algorithm, the cost function for candidate ranking is the Trajectory Error Percentage (TEP), calculated as the percentage of the ratio of Absolute Trajectory Error (ATE) [Sturm et al 2012] to the total trajectory length (L):

$$TEP = \frac{ATE}{L} \times 100\%. \tag{3.24}$$

The TEP effectively measures the per cent drift per metre of motion.

For the optimisation of the proposed method, the cost function is a custom metric referred to as Global Alignment and Smoothness Profile ($GASP$). This is calculated by taking the

weighted sum of the ATE per metre and the standard deviation of the translational Relative Pose Error (RPE), as defined in Equation (3.25):

$$\text{GASP} = w_{\text{drift}} \cdot \left(\frac{ATE}{L}\right) + w_{\text{smoothness}} \cdot \text{Translational RPE}. \tag{3.25}$$

The weights w_{drift} and $w_{\text{smoothness}}$ have been empirically tuned such that each term contributes approximately equally to the total cost. Because the weighted sum mixes quantities with different units, GASP serves as a relative performance index; its values are meaningful only when compared within the same experimental framework.

3.4 RESULTS

This section presents the empirical results from the evaluation of three localisation configurations: the baseline GLIM algorithm with default parameters, the refined GLIM algorithm with optimised parameters, and the proposed GNSS-augmented method. Each configuration was evaluated on five vineyard datasets. Performance is reported using Root Mean Square Error (RMSE), Absolute Trajectory Error (ATE), Relative Pose Error (RPE), and, for the proposed method, the Global Alignment and Smoothness Profile (GASP) metric defined in the methodology.

3.4.1 Baseline GLIM Performance

table 3.5 reports the performance of the baseline GLIM algorithm across the five datasets. The error percentage per metre ranges from 1.253% (Dataset 1) to 7.361% (Dataset 2). The 3D RMSE ranges from 8.035 m (Dataset 1) to 38.107 m (Dataset 3), and the ATE ranges from 6.605 m to 30.444 m. The maximum single-point 3D error observed is 80.339 m in Dataset 2.

3.4.2 Refined GLIM Performance

table 3.6 reports the results for the refined GLIM algorithm following parameter optimisation. The error percentage per metre ranges from 0.6081% (Dataset 1) to 3.6150% (Dataset 2). The 3D RMSE ranges from 3.4808 m (Dataset 1) to 25.1789 m (Dataset 2), and the ATE ranges from 3.2315 m to 12.6781 m.

3.4.3 Proposed Method Performance

table 3.7 reports the results for the proposed GNSS-augmented method. The error percentage per metre ranges from 0.0153% (Dataset 4) to 0.0501% (Dataset 2). The 3D RMSE ranges from 0.1097 m (Dataset 0) to 0.2280 m (Dataset 2), and the ATE ranges from 0.0811 m to 0.1741 m. The GASP metric, which combines drift rate and local pose smoothness, returned values between 0.0274 (Dataset 0) and 0.0679 (Dataset 2).

Table 3.5: Baseline GLIM performance over the five test datasets

Metrics	**Dataset 0**	**Dataset 1**	**Dataset 2**	**Dataset 3**	**Dataset 4**
RMSE x (m)	20.776	7.253	4.097	4.747	3.598
RMSE y (m)	3.730	2.766	3.391	4.596	4.889
RMSE z (m)	15.107	1.747	35.970	36.908	25.879
RMSE 3D (m)	26.990	8.035	36.755	38.107	27.767
ATE (m)	19.877	6.605	25.921	30.444	25.331
max error 3D (m)	47.892	17.766	80.339	69.878	56.729
error per meter (m)	0.0340	0.0130	0.0740	0.0490	0.0420
error percent per meter (%)	3.3580	1.2530	7.3610	4.8970	4.1910
RPE 1m translation RMSE	0.4697	0.2944	0.7632	1.0127	0.5356
RPE 1m translation mean	0.3100	0.1390	0.3210	0.3790	0.3170
RPE 1m translation std	0.4125	0.2602	0.6767	0.9384	0.4285
RPE 1m translation max	3.4500	2.8930	4.1800	15.7180	2.8810
RPE 10m translation RMSE	2.3790	1.5040	3.3400	3.3230	2.5920
RPE 10m translation mean	2.2680	1.1230	2.8110	3.0490	2.4040
RPE 10m translation std	0.7134	1.0076	1.8240	1.4120	0.9390
RPE 10m translation max	6.6530	7.9020	11.2070	16.2210	7.0210

Table 3.6: Refined GLIM performance over the five datasets

Metrics	**Dataset 0**	**Dataset 1**	**Dataset 2**	**Dataset 3**	**Dataset 4**
RMSE x (m)	6.3487	2.5059	3.6146	3.5669	1.1221
RMSE y (m)	3.2186	2.3727	24.9149	3.4499	2.6844
RMSE z (m)	0.6987	0.4545	0.3994	1.0413	2.1387
RMSE 3D (m)	7.1522	3.4808	25.1789	5.0705	3.6110
ATE (m)	5.9190	3.2315	12.6781	4.6763	3.3144
max error 3D (m)	14.3770	5.5867	71.0506	13.0417	5.8597
error per meter (m)	0.0087	0.0061	0.0362	0.0075	0.0067
error percent per meter (%)	0.8735	0.6081	3.6150	0.7499	0.6680
RPE 1m translation RMSE	0.3907	0.2609	0.5692	0.8317	0.3689
RPE 1m translation mean	0.1471	0.0899	0.3037	0.1133	0.1358
RPE 1m translation std	0.3619	0.2449	0.4814	0.8239	0.3430
RPE 1m translation max	3.1190	2.8235	2.3976	15.1426	2.7154
RPE 10m translation RMSE	1.2280	1.5267	4.6931	2.6912	1.4290
RPE 10m translation mean	0.7309	0.5967	2.6938	0.8912	0.5826
RPE 10m translation std	0.9868	1.4052	3.8430	2.5394	1.3048
RPE 10m translation max	5.7503	7.7271	10.7189	15.4780	6.6573

Table 3.7: Proposed method performance over the 5 test datasets

Metrics	**Dataset 0**	**Dataset 1**	**Dataset 2**	**Dataset 3**	**Dataset 4**
RMSE x (m)	0.0756	0.0528	0.0909	0.0528	0.0728
RMSE y (m)	0.0645	0.0906	0.2003	0.0906	0.0789
RMSE z (m)	0.0465	0.0347	0.0599	0.0347	0.0560
RMSE 3D (m)	0.1097	0.1105	0.2280	0.1105	0.1211
ATE (m)	0.0846	0.0811	0.1741	0.0811	0.1031
max error 3D (m)	0.7363	0.8963	1.0246	0.8963	0.7056
error per meter (m)	0.0002	0.0002	0.0005	0.0002	0.0002
error percent per meter (%)	0.0171	0.0154	0.0501	0.0154	0.0153
gasp	0.0274	0.0290	0.0679	0.0290	0.0429
RPE 1m translation RMSE	0.0270	0.0308	0.0556	0.0308	0.0444
RPE 1m translation mean	0.0175	0.0206	0.0282	0.0206	0.0249
RPE 1m translation std	0.0205	0.0229	0.0479	0.0229	0.0368
RPE 1m translation max	0.2001	0.4906	0.6992	0.4906	0.6462
RPE 10m translation RMSE	0.0649	0.0936	0.1772	0.0936	0.0769
RPE 10m translation mean	0.0433	0.0613	0.0987	0.0613	0.0573
RPE 10m translation std	0.0482	0.0708	0.1471	0.0708	0.0512
RPE 10m translation max	0.3799	0.7620	0.8566	0.7620	0.5377

3.4.4 Comparative Summary

table 3.8 presents a side-by-side comparison of the three methods on three key metrics: error percentage per metre, RPE 1 m translation standard deviation, and error per metre. The lowest value for each metric–dataset combination is indicated in bold.

3.4.5 GNSS Factor Utilisation

table 3.9 reports the number of GNSS factors incorporated into the proposed method's factor graph relative to the total number of available GNSS measurements. Across the five datasets, the proportion of GNSS points retained after outlier rejection and incorporated as factors ranges from 0.65% (Dataset 2) to 0.78% (Dataset 3).

3.4.6 Parameter Optimisation Progression

fig. 3.1 shows the distribution of per-iteration errors across the four optimisation stages. fig. 3.1a plots the error percentage per metre for the baseline GLIM tuning, and fig. 3.1b plots the GASP metric for the proposed method tuning. Both plots use a logarithmic horizontal axis.

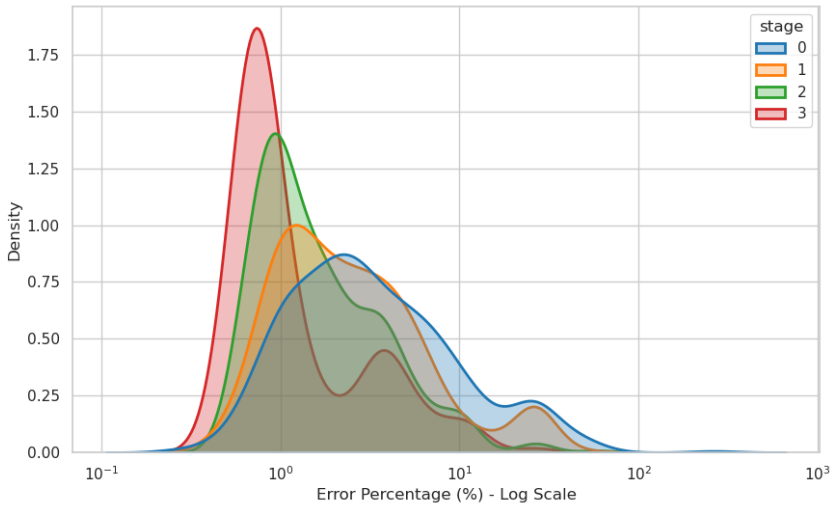
fig. 3.2 plots the best consensus error at the end of each optimisation stage. fig. 3.2a tracks the error percentage for the baseline GLIM, and fig. 3.2b tracks the GASP metric for the proposed method. The baseline error decreases monotonically across the four stages; the proposed method's GASP score increases from stage 0 to stage 3.

Table 3.8: Comparison Table of the Key Metrics (Best results in bold)

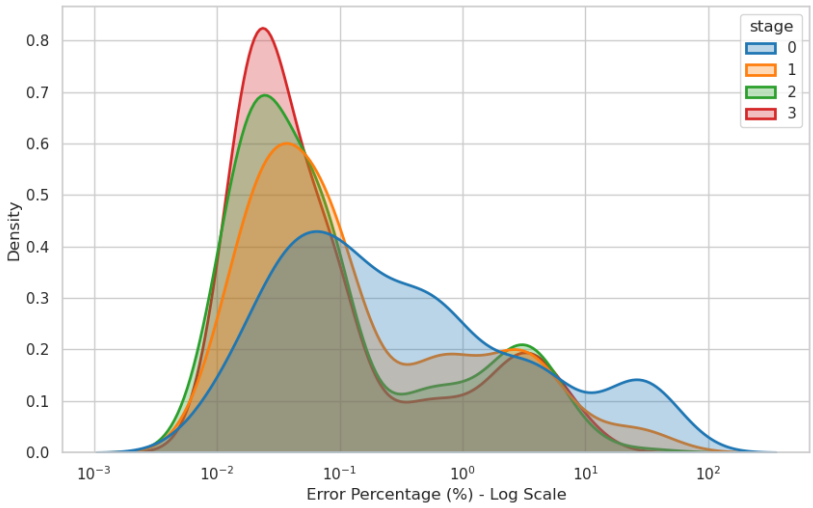
Metric	**Method**	**Dataset 0**	**Dataset 1**	**Dataset 2**	**Dataset 3**	**Dataset 4**
error percent per meter (%)	Baseline GLIM	3.3580	1.2530	7.3610	4.8970	4.1910
	Refined GLIM	0.8735	0.6081	3.6150	0.7499	0.6680
	Proposed Method	**0.0171**	**0.0154**	**0.0501**	**0.0154**	**0.0153**
RPE 1m translation std	Baseline GLIM	0.4125	0.2602	0.6767	0.9384	0.4285
	Refined GLIM	0.3619	0.2449	0.4814	0.8239	0.3430
	Proposed Method	**0.0205**	**0.0229**	**0.0479**	**0.0229**	**0.0368**
error per meter (m)	Baseline GLIM	0.0340	0.0130	0.0740	0.0490	0.0420
	Refined GLIM	0.0087	0.0061	0.0362	0.0075	0.0067
	Proposed Method	**0.0002**	**0.0002**	**0.0005**	**0.0002**	**0.0002**

Table 3.9: Statistics on the number of GNSS factors used for the proposed method

Metric	**Dataset 0**	**Dataset 1**	**Dataset 2**	**Dataset 3**	**Dataset 4**
Number of factors	93	73	47	86	69
Number of GNSS points	13461	10090	7236	11071	9241
Percent of GNSS points used	0.69	0.72	0.65	0.78	0.75

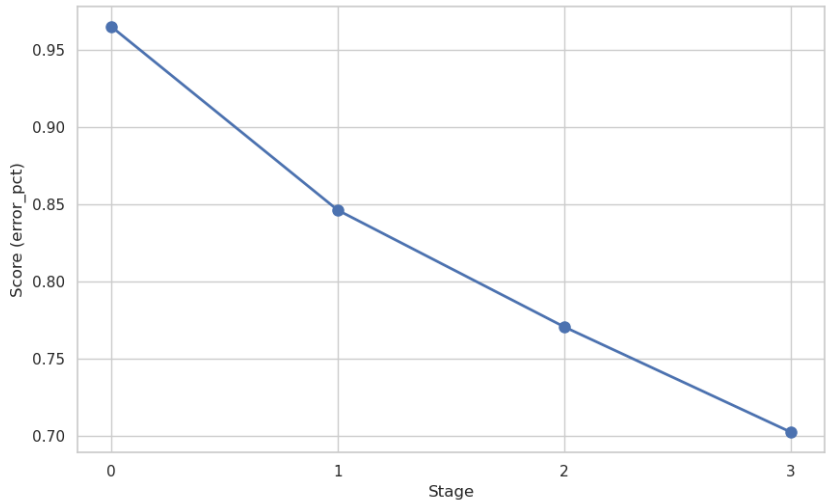


(a) Baseline glim tuning.

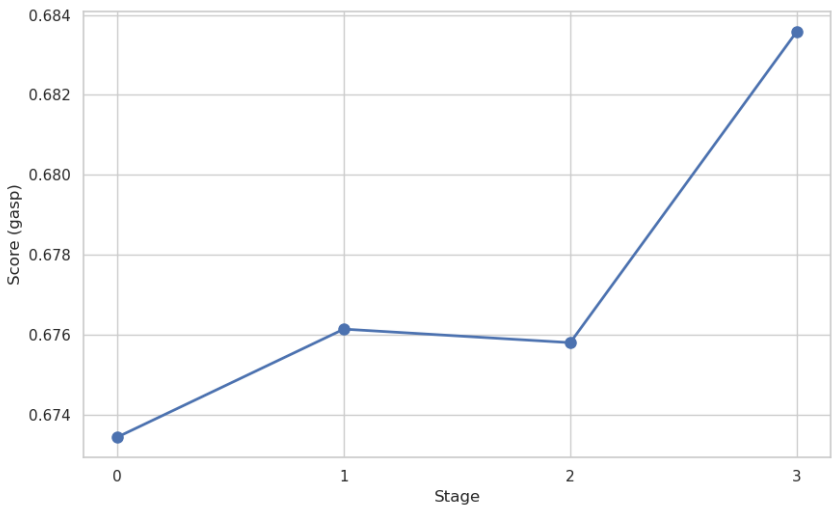


(b) Proposed method tuning.

Figure 3.1: Error distribution (log scale) over optimisation stages for baseline tuning (error = error percentage) and proposed method tuning (error = GASP).



(a) Baseline GLIM error percentage.

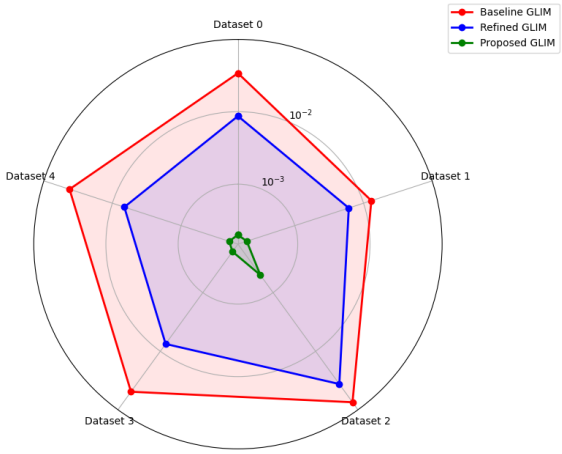


(b) Proposed Method GASP error.

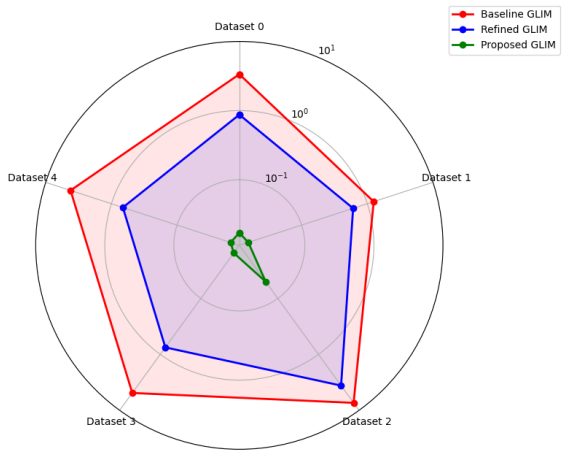
Figure 3.2: Consensus error over optimisation stages for baseline tuning (error = error percentage) and proposed method tuning (error = GASP).

3.4.7 Multi-Metric Radar Comparison

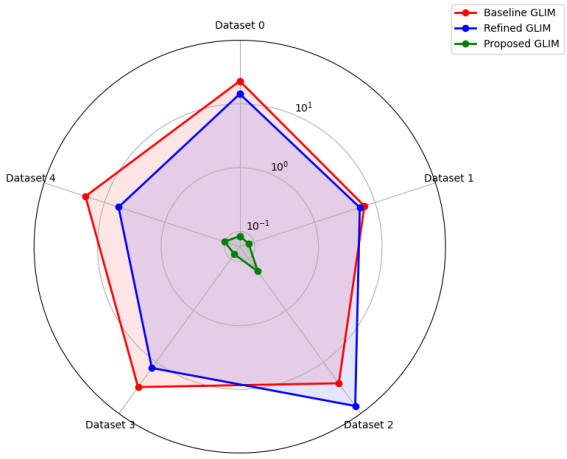
fig. 3.3 presents four radar plots comparing the baseline, refined, and proposed methods across all five datasets. The four metrics displayed are error per metre, error percentage per metre, ATE, and RPE 1 m translation standard deviation. Radial axes are plotted on a logarithmic scale.



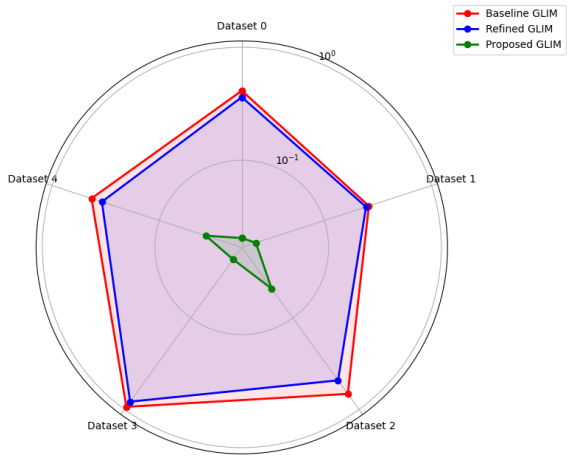
(a) Error per meter (m)



(b) Error percent per meter.



(c) ATE error (m).



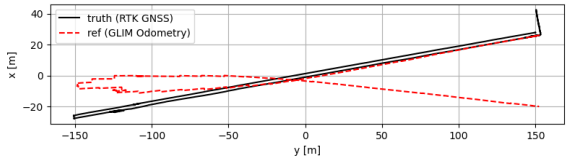
(d) RPE (translation only) standard deviation.

Figure 3.3: Spider plot comparisons of the glim baseline, refined and proposed methods for various key performance metrics.

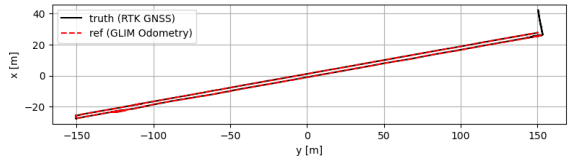
3.4.8 Trajectory Visualisation

fig. 3.4 presents the planar (x–y) trajectories for each dataset. The left column shows the refined GLIM output and the right column shows the proposed method output, both plotted against the RTK GNSS ground truth.

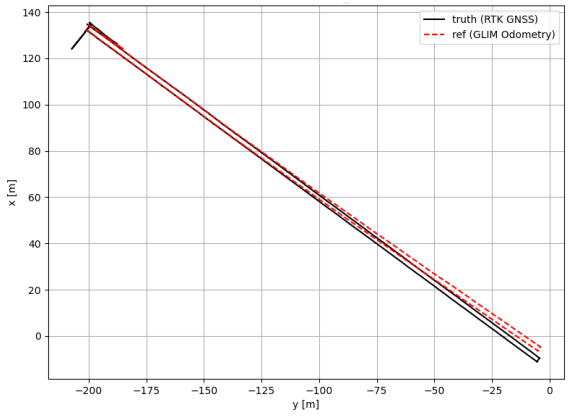
fig. 3.5 presents the altitude (z-axis) trajectories for each dataset in the same layout, with the refined GLIM estimates on the left and the proposed method estimates on the right, both compared against RTK GNSS ground truth.



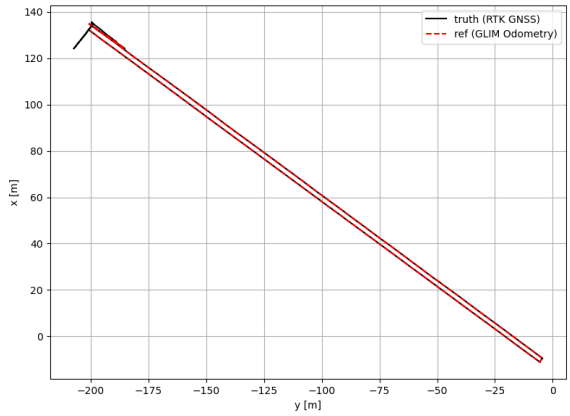
(a) Baseline | Dataset 0



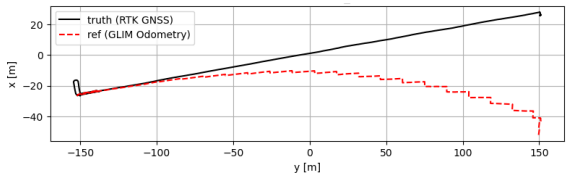
(b) Proposed | Dataset 0



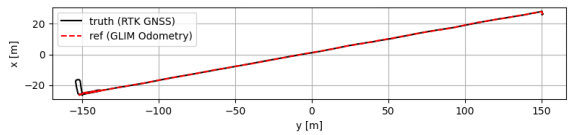
(c) Baseline | Dataset 1



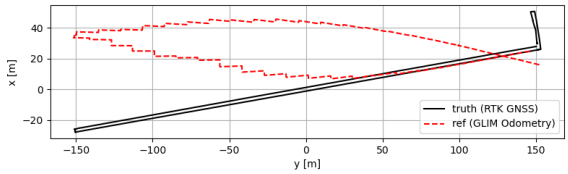
(d) Proposed | Dataset 1



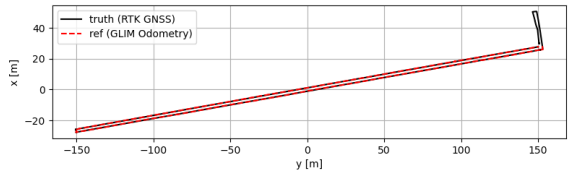
(e) Baseline | Dataset 2



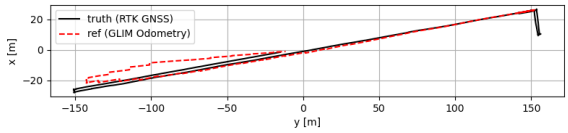
(f) Proposed | Dataset 2



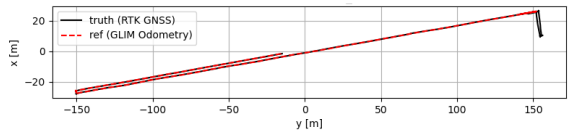
(g) Baseline | Dataset 3



(h) Proposed | Dataset 3

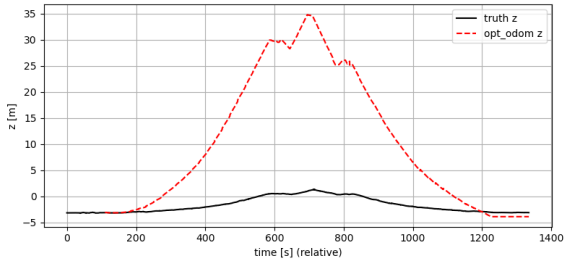


(i) Baseline | Dataset 4

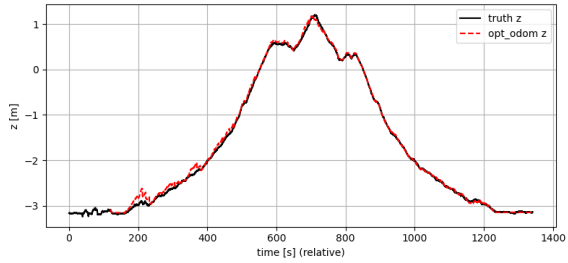


(j) Proposed | Dataset 4

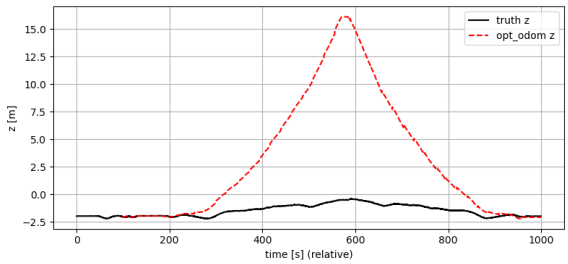
Figure 3.4: Trajectories in XY plane from baseline GLIM and the proposed method.



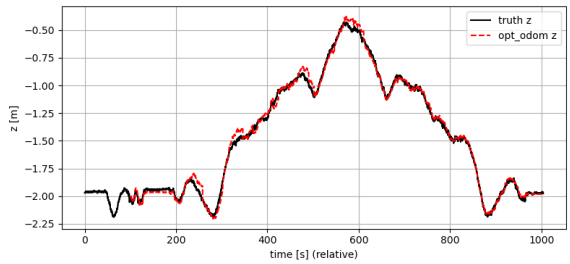
(a) Baseline | Dataset 0



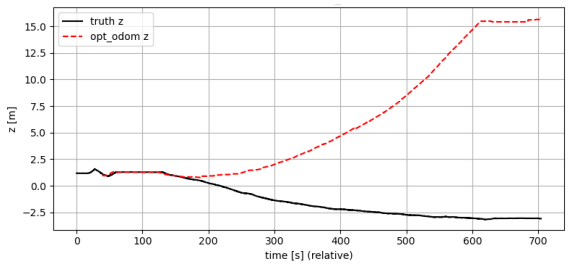
(b) Proposed | Dataset 0



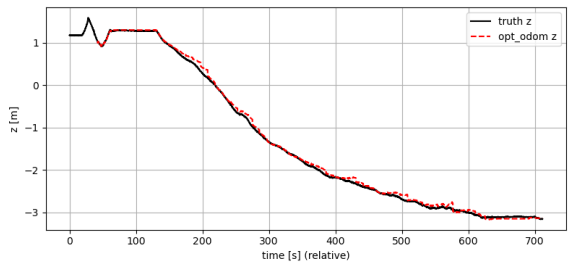
(c) Baseline | Dataset 1



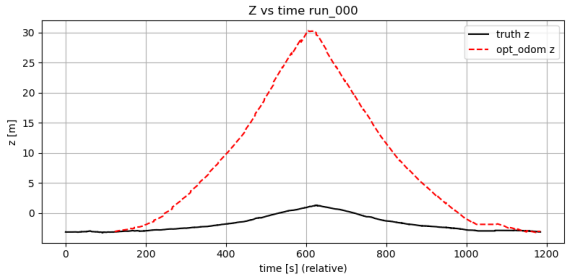
(d) Proposed | Dataset 1



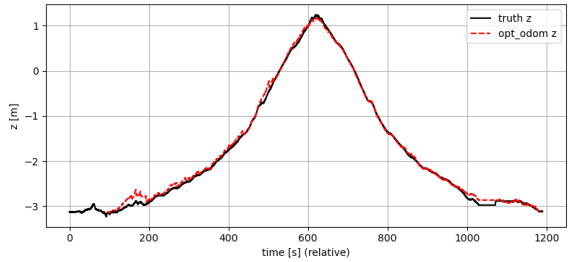
(e) Baseline | Dataset 2



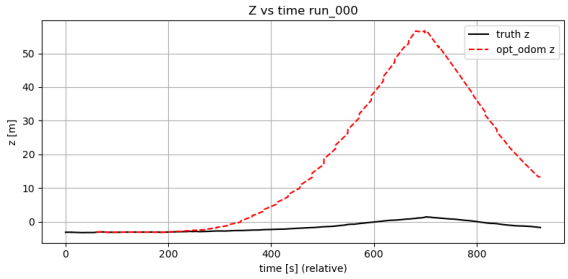
(f) Proposed | Dataset 2



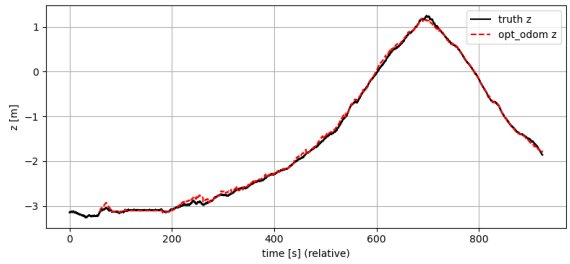
(g) Baseline | Dataset 3



(h) Proposed | Dataset 3



(i) Baseline | Dataset 4



(j) Proposed | Dataset 4

Figure 3.5: Trajectories in Z plane from baseline GLIM and the proposed method.

3.5 DISCUSSION

3.5.1 Baseline GLIM Performance Analysis

The evaluation of the baseline GLIM algorithm revealed two critical failure modes that fundamentally compromised its localisation reliability: systemic vertical drift and a catastrophic "point of no return" in planar estimation.

3.5.1.1 Vertical Drift and Altitude Recovery

The baseline algorithm exhibited a pronounced systemic drift in the positive altitude (z-axis) direction. This phenomenon is clearly illustrated in Dataset 4 (Figure 3.5g), where the vehicle traverses a linear trajectory down a vineyard row. Although the actual environment features a modest elevation gain of approximately 5 m, the baseline GLIM algorithm accumulated a significant vertical error, culminating in nearly 30 m of drift by the end of the row.

Interestingly, this drift demonstrated a self-correcting property during the return traversal. As the vehicle moved in the opposite direction, the vertical error decreased at a rate symmetrical to its initial accumulation, eventually reconverging with the ground truth data. This behavior suggests that the initial drift was effectively encoded into the generated LiDAR map. During the return pass, the scan-matching process was constrained by this previously mapped geometry. Consequently, the odometry "undid" the accumulated drift by aligning the current scans with the (erroneous) elevation of the existing map.

Notably, this vertical distortion of the map did not appear to hinder subsequent scan matching or loop closure. The algorithm remained capable of registering scans against a map with significant Z-axis distortion. This resilience may be a byproduct of the specific experimental geometry—characterized by repetitive A-B linear paths with minimal lateral movement—which allows the scan matcher to maintain a high correlation in the X-Y plane despite vertical inconsistencies.

3.5.1.2 Planar Failure and the 'Point of No Return'

In contrast to the recoverable nature of vertical drift, errors in the X-Y plane frequently led to catastrophic and irrecoverable failure. This "point of no return" is most evident in Figure 3.4a, where the baseline algorithm maintains high fidelity for the first half of the traversal before experiencing a sudden, exponential increase in error.

At this critical threshold, the accumulated planar drift distorts the LiDAR map to an extent that subsequent scan matching becomes fundamentally unreliable. Once in this failure state, the trajectory becomes increasingly jagged and stochastic, likely caused by false-positive scan matches that introduce large, discontinuous jumps in the pose estimation.

This failure mode is particularly prevalent in the structured and repetitive environments of vineyard rows. Small angular or lateral drifts cause parallel rows in the LiDAR map to diverge or overlap incorrectly. Once this spatial distortion exceeds the scan matcher's convergence radius, the algorithm may incorrectly register the vehicle's current position to an adjacent row. This

"row-skipping" effect creates a feedback loop of map corruption, where each incorrect match further distorts the global map, rendering recovery unlikely for the remainder of the trajectory.

3.5.2 Refined GLIM Performance

The transition from the baseline GLIM algorithm to the refined (parameterised) version demonstrates a significant enhancement in localisation stability and drift suppression. While the baseline algorithm frequently succumbed to the failure modes described in previous sections, the refined version exhibits a more controlled error growth profile.

3.5.2.1 Magnitude of Error Reduction

The most prominent improvement is observed in the *error percent per metre* metric. In the baseline configuration, Dataset 2 exhibited a peak error of 7.3610%, whereas the refined version reduced this to 3.6150%, a reduction of approximately 50%. In more favourable conditions, such as Dataset 4, the improvement was even more dramatic: the error dropped from 4.1910% in the baseline to a significantly lower 0.6680% in the refined version. This indicates that parameter optimisation effectively mitigates the "point of no return" by maintaining better geometric alignment over longer durations.

3.5.2.2 Relative Pose Stability

The *RPE 1 m translation standard deviation* provides insight into the "smoothness" or "jitter" of the trajectory. While the baseline GLIM suffered from jagged trajectories around points of failure – peaking at a standard deviation of 0.9384 in Dataset 3 – the refined GLIM consistently lowered this jitter across all datasets. For instance, in Dataset 3, the refined version achieved an RPE standard deviation of 0.8239. Although this remains the highest in the refined category, the overall reduction suggests that the optimised parameters facilitate more consistent scan-matching, reducing the frequency of the "jumps" caused by false-positive matches in repetitive environments.

3.5.2.3 Drift Suppression per Meter

The *error per metre (m)* metric further reinforces these findings. The refined GLIM consistently achieved sub-centimetre drift per metre in datasets where the baseline was losing several centimetres (e.g., Dataset 0 dropping from 0.0340 m to 0.0087 m). This suggests that the refinement process successfully tuned the fusion weights and scan-matching thresholds to better suit the specific LiDAR characteristics and environment structure, delaying the onset of the catastrophic failures observed in the baseline.

Table 3.10: Comparative Performance: Baseline vs. Refined GLIM

Metric	Dataset 0	Dataset 1	Dataset 2	Dataset 3	Dataset 4
Error % Reduction	73.9%	51.4%	50.8%	84.6%	84.0%
RPE Std Improvement	12.2%	5.8%	28.8%	12.2%	19.9%

3.5.3 Proposed Method: GNSS-Augmented Localisation

The integration of GNSS measurements into the GLIM factor graph constitutes a fundamental architectural change: by introducing a source of absolute positional observability absent from both the baseline and refined GLIM configurations, it eliminates the unbounded drift inherent in scan-matching-only odometry. The resulting improvement of approximately two orders of magnitude in localisation accuracy relative to the no-GNSS baseline is consistent with this design intent, validating the fusion architecture rather than representing an unexpected outcome. The proposed method's error percentage per metre ranges from 0.0153% (Dataset 4) to 0.0501% (Dataset 2), compared to the refined GLIM's range of 0.6081% to 3.6150%. This reduction is consistent across all datasets, with no observable catastrophic failure modes analogous to those seen in earlier iterations.

3.5.3.1 GNSS Integration and Global Constraint

The performance improvement stems fundamentally from the incorporation of GNSS-derived positional constraints within the factor graph optimisation. Whereas GLIM operates exclusively on scan-matching odometry, the proposed method augments this local relative information with periodic absolute position measurements from the GNSS receiver. This integration serves two critical functions: (1) it provides global observability that prevents unbounded drift accumulation, and (2) it allows the factor graph optimiser to "correct" local odometric errors retroactively across the entire trajectory window.

The effectiveness of this approach is evident in comparing error accumulation profiles. In the refined GLIM, even with parameter optimisation, the error percent per metre remains linear or superlinear across the traversal. By contrast, the proposed method maintains sub-centimetre per-metre errors throughout, suggesting that the global constraints are sufficiently frequent and accurate to prevent drift from establishing itself in the first place.

This result is further reinforced by the GNSS factor utilisation statistics (Table 3.7 in the results section): across all datasets, between 0.65–0.78% of all available GNSS datapoints were injected into the factor graph following outlier rejection. This sparsity is itself a key finding of this work. A primary motivation for the proposed architecture was to reduce operational dependence on RTK-GNSS, which is susceptible to canopy occlusion, multipath interference, weather conditions, and infrastructure cost in agricultural deployments. The results demonstrate that the vast majority of localisation accuracy is recovered through the LiDAR-inertial odometry backbone, with GNSS contributing only infrequent global anchors to prevent long-

term drift accumulation. Achieving two orders of magnitude improvement over the no-GNSS baseline at sub-percent GNSS utilisation confirms that dense or continuous RTK coverage is not a prerequisite for precise trajectory estimation in this architecture.

3.5.3.2 Dataset-Specific Performance Variability

Despite the strong performance of the proposed method across all datasets, performance does vary across datasets in a pattern that reveals environmental factors influencing localisation quality. Dataset 2 exhibits the highest error metrics (0.0501% error per metre, RMSE y of 0.2003 m, ATE of 0.1741 m), whereas Datasets 1 and 3 achieve the lowest errors (0.0154% error per metre). This variability is instructive rather than concerning, as the absolute error magnitudes remain within acceptable bounds for agricultural manipulation.

The degradation in Dataset 2 is likely attributable to environmental conditions rather than algorithmic limitations. Visual inspection of the traversal pattern and GNSS utilisation (0.65% of available GNSS datapoints injected into the factor graph, the lowest across all datasets) suggests that portions of the trajectory experienced reduced GNSS signal quality or multi-path interference. The geometric consistency-based outlier rejection, whilst effective at filtering genuinely erroneous points, cannot recover lost signal availability. However, even under these challenging conditions, the localisation error remained controlled at 0.0501%, indicating robustness to transient GNSS degradation.

3.5.3.3 Relative Pose Error and Short-Horizon Consistency

The Relative Pose Error (RPE) metrics, which assess the consistency of local pose increments, provide insight into frame-to-frame localisation smoothness. The proposed method achieves RPE 1 m translation standard deviations ranging from 0.0205 to 0.0479 m across all datasets, compared to the refined GLIM's 0.2449 to 0.8239 m. This two-to-tenfold reduction in local pose jitter indicates that the global constraints are not merely correcting large-scale drift, but are improving the consistency of short-horizon relative measurements.

This improvement likely arises from the factor graph's ability to reason about odometric uncertainty jointly with global constraints. When the optimiser encounters a region where local odometry is inconsistent with global GNSS observations, it propagates the correction smoothly across the pose graph rather than applying isolated jump corrections. This smoothness is essential for downstream applications such as robotic arm control, where abrupt discontinuities in perceived position could cause trajectory tracking errors.

3.5.3.4 GASP Metric Behaviour and Optimisation Dynamics

A notable feature of the results is a slight upward trend in the GASP (Global Alignment and Smoothness Profile) metric for the single best-performing run across the four parameter optimisation stages. Unlike the baseline GLIM error percentage, which exhibits clear convergence and monotonic improvement, the proposed method's best-run GASP metric increases marginally from stage one to stage four. This trend is attributable to the stochastic nature of the Latin

Hypercube parameter search rather than to any systematic degradation: the differences between near-optimal configurations are negligible, and stage one already identifies a configuration that fully exploits the GNSS integration architecture. Subsequent stages sample nearby regions of the parameter space and, by chance, return configurations that score marginally worse on GASP, without producing any material change in trajectory accuracy. The magnitude of the variation is small enough that it falls within the expected noise of the pseudo-random search process.

Despite this marginal decline in peak performance, the overall distribution of GASP metric values improves significantly across the four stages. As visible in fig. 3.1b, the error distributions shift progressively leftward over successive stages, indicating an improvement in typical performance. Concurrently, the distributions become taller and narrower — that is, the standard deviation decreases — reflecting a marked increase in the reliability and consistency of results. In contrast, the stage zero distribution is comparatively flat and wide, indicating that results at that stage were highly variable and unpredictable. This evolution demonstrates that the optimisation process is not merely identifying a single lucky configuration, but is systematically converging the parameter space towards a regime in which good performance is reproducible. The trade-off between peak and median performance is therefore a natural consequence of this convergence: as the search focuses on consistently performant regions, extreme but unreliable high-performing outliers are replaced by a tighter cluster of reliably good solutions.

This observation underscores an important principle: no single metric fully captures the multidimensional nature of SLAM performance. GASP is most informative when comparing qualitatively different configurations (e.g., baseline vs. GNSS-augmented); within a near-optimal neighbourhood, trajectory accuracy metrics such as ATE and RPE are more reliable discriminators.

3.5.4 Comparative Analysis and Mechanisms of Improvement

Examining the three approaches in concert reveals a clear hierarchy of performance and elucidates the mechanisms driving improvements at each stage.

3.5.4.1 Parametrisation vs. Sensor Fusion

The transition from baseline to refined GLIM demonstrates that parameter tuning can reduce error magnitude significantly (50–85% reduction in error percentage). However, parameter optimisation alone does not eliminate the fundamental instability of scan-matching-only odometry in repetitive vineyard environments. The refined GLIM still exhibits occasional trajectory inconsistencies and elevated RPE metrics relative to the proposed method.

By contrast, the transition from refined GLIM to the GNSS-augmented method represents a qualitative shift in localisation architecture. Rather than optimising the tuning parameters of a single sensor modality, the proposed method introduces a fundamentally independent source of global position information. This architectural change achieves improvements that parametrisation alone cannot reach. The error reduction from refined GLIM to the proposed method (∼50–98%) is larger and more consistent than the improvement from baseline to re-

fined (∼50–85%), and the proposed method exhibits no dataset in which performance degrades significantly.

3.5.4.2 Role of Outlier Rejection

The geometric consistency-based outlier rejection procedure is a critical enabling factor in this result. The procedure admits only those GNSS measurements that satisfy pairwise distance-preservation constraints across the candidate window, rejecting the large majority of available raw measurements before factor graph integration. This aggressive filtering is precisely what makes the sparse utilisation viable: by ensuring that every incorporated constraint is geometrically consistent, the procedure prevents a single erroneous global measurement from corrupting the factor graph. The architecture therefore achieves the desired property of GNSS-frugal localisation — using the global sensor as a rare but reliable anchor rather than a continuous dependency. This deterministic filtering approach, based on rigid-body distance preservation, is well-suited to the repetitive vineyard environment, where probabilistic alternatives such as RANSAC would require outlier distribution assumptions that may not hold in structured agricultural settings.

The robustness of this design is confirmed by Dataset 2, where GNSS factor utilisation falls to its lowest observed value of 0.65%, yet localisation error remains controlled at 0.0501% per metre. This indicates that each geometrically verified constraint carries sufficient positional information to anchor the trajectory even when GNSS availability is degraded.

3.5.4.3 Geometric Quality Assessment and Kabsch Stability

The planar geometry quality assessment phase, which enforces minimum requirements on translational baseline, spread ratio, rotational excitation, and condition number, ensures that the Kabsch algorithm operates in a numerically stable regime. This preprocessing step prevents ill-conditioned covariance matrices that could yield biased or unstable alignment solutions.

The quintet of measurement from Datasets 0–4 all satisfy the geometry quality thresholds, which explains the consistency of global alignment across the evaluation. In datasets with weakly excited geometry (e.g., highly linear traversals with minimal rotation), this step would conservatively defer global alignment until sufficient geometric excitation is accumulated, preventing premature or inaccurate initialisation.

3.5.5 Implications for Vineyard Robotics

The proposed method's consistent sub-centimetre accuracy has direct implications for robotic pruning feasibility. Pruning tasks require localisation accuracy sufficient to position the robotic arm within reach of target vine structures. With typical vine stems ranging from 5–15 mm in diameter and clusters requiring workspace accessibility, the proposed method's maximum absolute trajectory error of approximately 0.8–1.0 m (observed across all datasets) represents an acceptable error bound. This error magnitude can be accommodated through local perception-based arm guidance or online trajectory adjustment based on close-range stereo or depth sensing.

Furthermore, the low local pose jitter (RPE 1 m standard deviation of 0.0205–0.0479 m) is important for smooth arm trajectory execution. Abrupt jumps in perceived position, typical of the baseline and refined GLIM approaches, would cause tracking servo errors and potentially jerky or unsafe arm motion. The proposed method's smoothness mitigates this risk.

However, a critical limitation remains unaddressed by the current evaluation: all experiments were conducted at slow traversal speeds (approximately 0.5 m/s) with ample GNSS signal availability. Real vineyard deployments will encounter variable canopy occlusion, faster navigation speeds during inter-row transit, and periods of GNSS dropout beneath dense foliage. The robustness of the method under these challenging real-world conditions has not yet been empirically demonstrated. Chapter 4 will address this limitation by evaluating precise alignment techniques that can operate with minimal or absent GNSS input, using instead local dense perception and pre-existing 3D models to constrain pose estimation.

3.6 SUMMARY

This chapter presented the design and implementation of a LiDAR-centric SLAM pipeline for GNSS-degraded operation and evaluated three configurations: the baseline GLIM system, a refined parameterised variant, and the proposed method. The results across five datasets showed that parameter refinement reduced drift and improved trajectory smoothness relative to the baseline, while the proposed method achieved the lowest errors across the reported metrics. The discussion highlighted characteristic failure modes in structured vineyard environments and the trade-offs between global consistency and computational cost.

Chapter 4

PRECISE ALIGNMENT

4.1 INTRODUCTION

The preceding chapter established a LiDAR-inertial SLAM pipeline capable of operating without continuous GNSS availability, providing robust trajectory estimation and dense environmental mapping across vineyard rows. While this system delivers the metre-scale localisation required for autonomous navigation between vines, it does not, by itself, provide the sub-centimetre spatial accuracy necessary for robotic pruning. When a manipulator arm must reach a specific cane at a specific cut-point, the tolerance on positional error is far smaller than what global SLAM can guarantee. This chapter addresses that gap through a second-stage localisation refinement step that precisely aligns the robot's perception of its environment with a pre-built high-fidelity vine model.

4.1.1 System Context

Central to the pruning workflow described in this thesis is a 3D Gaussian Splatting (3DGS) model of each vine, constructed offline from imagery collected during a prior survey pass. The reconstruction pipeline applies Structure from Motion (SfM) photogrammetry to these images, recovering a dense 3D point cloud and camera trajectory via bundle adjustment. The resulting point cloud forms the geometric backbone of the 3DGS model and is registered within a global Local East-North-Up (ENU) reference frame. Because SfM jointly optimises camera poses and 3D structure through reprojection minimisation, the reconstructed geometry is internally consistent and, once georeferenced, constitutes a reliable proxy for the true physical vine geometry. All downstream tasks — cut-point detection, path planning for the robotic arm, and collision avoidance — are computed against this 3DGS model in simulation. For these simulated operations to correspond faithfully to the real world, the robot's pose relative to the simulated vine (i.e. the 3DGS reconstruction) must match its true pose relative to the physical vine.

When the robot arrives at a target vine during an autonomous pruning mission, its pose is known only to the accuracy of the SLAM system described in chapter 3. This accuracy is sufficient for navigation but insufficient for manipulation. The precise-alignment stage presented here corrects the residual misalignment by registering a real-time point cloud of the physical vine, acquired at the site, against the SfM point cloud extracted from the 3DGS model. The resulting

rigid-body transformation is used to update the robot's estimated pose, thereby aligning the simulated and physical scenes before any pruning motion is executed.

4.1.2 Approach Overview

The precise-alignment pipeline proceeds through the following stages:

1. The robot halts in front of the target vine and acquires a dense point cloud of the vine structure. Two sensing modalities are evaluated in this chapter: depth point clouds generated from a stereo camera, and dense maps built from a spinning LiDAR sensor.

2. The acquired point cloud is passed through a preprocessing pipeline that filters noise, removes the ground plane and non-structural clutter, and voxel-downsamples the data to a resolution suitable for registration.

3. The preprocessed source cloud is registered against the target cloud derived from the SfM reconstruction of the 3DGS model using the Iterative Closest Point (ICP) algorithm [Besl and McKay 1992]. ICP iteratively solves for the rigid transformation — a rotation $\mathbf{R} \in SO(3)$ and translation $\mathbf{t} \in \mathbb{R}^3$ — that minimises the sum of squared distances between corresponding point pairs in the two clouds.

4. The recovered transformation is applied to update the robot's internal localisation, correcting its estimated pose relative to the vine. This updated pose is consumed by the arm path planner, ensuring that simulated cut-points correspond to their physical counterparts.

Each stage of this pipeline draws on a distinct body of prior work. This chapter is structured as follows. Section 4.2 reviews prior work on point cloud registration for agricultural and robotic applications, with particular attention to ICP variants and vine-specific sensing modalities. Section 4.3 details the full data processing and registration pipeline, including sensor configuration, preprocessing steps, and ICP parameterisation. Section 4.4 presents quantitative and qualitative results for both sensing modalities across a set of vineyard test cases. Section 4.5 discusses the relative merits and failure modes of each approach, and section 4.6 summarises the key findings.

4.2 RELATED WORK

The precise localisation of a robotic platform within a vineyard environment spans three complementary technical domains. First, the robot must generate high-fidelity, metrically scaled three-dimensional representations of the immediate vine environment from its onboard sensors. Second, a global structural reference map must already exist against which those local observations can be compared; in the system described in this thesis, this reference is a Structure from Motion (SfM) point cloud extracted from a 3D Gaussian Splatting model, though LiDAR-derived maps serve the same purpose in other systems. Third, reliable algorithms must compute the rigid transformation that aligns the local capture to the global frame, recovering

the robot's pose with centimetric accuracy. The following review surveys each domain in turn: proximal three-dimensional sensing via stereo, RGB-D, and LiDAR modalities; the construction of dense reference maps from LiDAR and photogrammetric pipelines; and the class of point cloud registration algorithms built upon the Iterative Closest Point (ICP) framework.

4.2.1 Stereo Depth Imaging

Stereo and RGB-D imaging systems are widely used to generate proximal three-dimensional representations of individual plants. Unlike LiDAR, these systems are compact and inexpensive, making them practical for mounting on a field robotics platform. Depth is recovered by computing the disparity d between a spatially separated pair of images, related to metric depth Z by the pinhole model:

$$Z = \frac{f \cdot B}{d} \tag{4.1}$$

where f is the focal length and B is the baseline between the two image sensors. The principal challenge in vineyard settings arises from the self-occluding geometry of vine canopies and the variability of outdoor illumination.

Early work by Klodt et al. [Klodt et al 2015] established that passive stereo cameras are sufficient for quantitative vine characterisation. Their dense reconstruction pipeline generated depth maps from paired RGB images, enabling background subtraction and 3D estimation of leaf area and fruit-to-leaf ratios directly in the field, without requiring a controlled background or calibration rig. This demonstrated that the stereo approach can yield reliable vine structure measurements under realistic outdoor conditions. Parr et al. [Parr et al 2022] subsequently benchmarked five depth sensor modalities on grape clusters under both laboratory and open-field conditions, comparing structured light, active-infrared (IR) stereo, time-of-flight, and LiDAR-based sensors. Their results showed that the RealSense D415 active IR stereo sensor achieved approximately 2 mm error in controlled indoor conditions and remained functional in full sunlight, whereas time-of-flight and structured-light sensors exhibited larger measurement bias. These comparative findings provide an empirical basis for depth sensor selection in proximal vineyard tasks.

Modern acquisition pipelines have shifted towards offline Structure from Motion and Multi-View Stereo (SfM/MVS) processing of high-resolution images, rather than relying on the camera's integrated disparity engine. Wang et al. [Wang et al 2025a] proposed and validated a two-phase plant 3D reconstruction workflow in which SfM/MVS processing of images taken from six viewpoints generates high-fidelity, single-view point clouds free from the geometric distortion inherent to single-shot disparity computation. These partial clouds are then registered into a unified model via a coarse marker-based alignment followed by ICP fine refinement. Evaluated on two *Ilex* species, the method achieved coefficients of determination (R^2) exceeding 0.92 for plant height and crown width, validating its suitability for plant-level quantitative phenotyping. A related formulation by Tamura et al. [Tamura et al 2025] applied unsupervised deep learning to omnidirectional monocular imagery for 3D position estimation of grape berries in the wild. The model was trained using photometric reconstruction error and structural similarity (SSIM) as loss functions, enabling depth inference on the textureless surfaces typical of

tightly packed grape clusters—surfaces on which conventional feature-matching stereo typically fails.

More recently, foundation model approaches have been applied to stereo depth estimation with the goal of achieving reliable zero-shot generalisation across varied imaging conditions. FoundationStereo [Wen et al 2025] addresses the sim-to-real transfer problem by training on a curated dataset of one million photorealistic synthetic stereo pairs and employing a Side-Tuning Adapter (STA) to leverage monocular depth priors from DepthAnything V2; cost volumes are subsequently filtered via an Attentive Hybrid Cost Filtering (AHCF) module combining Axial-Planar Convolutions with a Disparity Transformer for long-range disparity reasoning. Evaluated on the Middlebury and ETH3D benchmarks, the model achieves state-of-the-art results and generalises well to in-the-wild data. In the present work, FoundationStereo is applied to imagery from a ZED Mini stereo camera to produce the depth maps from which the source point cloud is derived prior to registration.

Active RGB-D sensors, including the Intel RealSense series and the Microsoft Azure Kinect, use infrared projection to supply artificial texture in scenes that would be feature-poor under passive stereo alone. Miranda et al. [Miranda et al 2023] evaluated the Azure Kinect for in-field fruit size estimation and documented a critical limitation: direct sunlight saturates the IR projector, leading to systematic depth failures in open canopy conditions. This sensitivity to ambient illumination is a primary practical constraint for any deployment relying on active structured-light depth in outdoor vineyards.

Multi-view registration is a necessary extension when reconstructing structurally complex plants that cannot be captured adequately from a single viewpoint. Stumpe et al. [Stumpe et al 2025] addressed this challenge in the context of plant phenotyping, proposing a multimodal 3D image registration method that integrates time-of-flight depth data to mitigate the parallax and occlusion artefacts that arise when fusing images from different viewpoints and sensor types.

4.2.2 Dense LiDAR Mapping

The generation of a dense, georeferenced global map is a prerequisite for the relocalisation paradigm. Light Detection and Ranging (LiDAR)—an active, illumination-invariant modality that measures precise time-of-flight returns—has become the primary technology for constructing such reference maps in outdoor agricultural environments [Farhan et al 2024]. Unlike stereo imaging, LiDAR is unaffected by scene texture and performs consistently during the dormant season when vine architecture is reduced to bare wood, which is precisely when precision pruning operations are carried out.

Llorens et al. [Llorens et al 2011] demonstrated georeferenced tractor-mounted vineyard mapping by combining a SICK LiDAR profiler with a GPS receiver. By establishing the spatial correspondence between canopy return density and GPS coordinates at driving speed, they produced vineyard-scale canopy maps that were overlaid onto satellite imagery. This side-scanning acquisition paradigm—in which the sensor traverses the row and records multiple returns per vine—established the operational approach adopted by most subsequent ground-level systems. Moreno et al. [Moreno et al 2020] extended the concept with a downward-facing 2D time-of-

flight LiDAR mounted on an electric platform georeferenced via RTK-GNSS, generating high-density 3D point clouds of dormant vine rows. Applying α-shape fitting to extract branch volumes, they reported a linear correlation with actual dry biomass of $R^2 = 0.85$, confirming that LiDAR volumetrics provide a reliable proxy for agronomically relevant pruning wood measurements. A comparable mobile 3D LiDAR and IMU system was evaluated by Chakraborty et al. [Chakraborty et al 2019] for canopy mapping in fruit crops including grapevines; voxel-grid and convex-hull volume methods yielded a Pearson correlation of $r \approx 0.81$ with manual measurements, with the added capability of monitoring seasonal canopy development over multiple visits.

The integration of SLAM with LiDAR enables simultaneous pose tracking and map construction during robot traversal, eliminating the dependence on external GNSS in canopy-occluded or signal-denied environments. Tagarakis et al. [Tagarakis et al 2026] demonstrated SLAM-based LiDAR for generating digital maps of pruning wood biomass across vineyard parcels, illustrating how a pre-built structural map can directly support precision viticulture decision-making. Combining LiDAR structural data with complementary sensing modalities improves the semantic richness of the resulting maps. Pavol Jozef et al. [Pavol Jozef Šafárik University in Košice et al 2025] fused UAV hyperspectral imagery with LiDAR-derived canopy height models in the Tokaj vineyard region and found that spectral classification alone achieved only 48.9% producer accuracy for vine detection, whereas adding structural LiDAR features improved overall accuracy to 96%. The AgriLiRa4D benchmark dataset [Zhan et al 2026] extends multi-modal agricultural mapping to the aerial domain, providing synchronised 3D LiDAR, 4D radar, and IMU data from UAV flights over flat, hilly, and terraced farmland types. The dataset is specifically constructed to evaluate multi-sensor SLAM algorithms under the challenges characteristic of agricultural fields—including low-texture crops, repetitive geometric patterns, and dynamic vegetation—and defines a public benchmark for comparative evaluation.

Beyond its established role in large-scale mapping, LiDAR is equally applicable as a proximal sensing modality for capturing individual vine geometry at close range. When mounted on a robotic platform that halts adjacent to a target vine, a spinning 3D LiDAR sensor accumulates dense returns from the vine structure over a short acquisition period. The resulting point cloud, though captured from a limited set of viewpoints, can provide millimetre-resolution coverage of the exposed wood and cane geometry—a capability that is particularly valuable during the dormant season, when visual texture is sparse and passive stereo matching becomes unreliable. This proximal use of LiDAR forms one of the two sensing modalities evaluated in the present work.

A direct comparison of LiDAR and photogrammetric reconstruction on defoliated vines was conducted by Petrović et al. [Petrović et al 2022], who found that both terrestrial LiDAR and UAV photogrammetry reconstruct canopy geometry with comparable fidelity, each achieving regression against ground-truth geometry of $R \approx 0.9$. This equivalence indicates that dense SfM-generated point clouds are a viable alternative to direct LiDAR for constructing global reference maps, provided sufficient image overlap and resolution are maintained during acquisition. In the system described in this thesis, the global reference is an SfM-derived point cloud extracted from a 3D Gaussian Splatting model of each vine, and it is against this reference that the locally

acquired source cloud is registered.

Whether the global reference originates from LiDAR or from SfM photogrammetry, and whether the local observation is captured by a stereo camera or a spinning LiDAR, the final alignment step requires a common algorithmic framework for registering the two point clouds.

4.2.3 Point Cloud Registration

Point cloud registration is the core computational operation in the precise alignment problem: given a local point cloud captured by the robot's onboard sensor and a pre-built global reference map, the objective is to find the rigid transformation—comprising a rotation matrix $\mathbf{R} \in SO(3)$ and a translation vector $\mathbf{t} \in \mathbb{R}^3$—that minimises the geometric discrepancy between the two sets. The canonical formulation, proposed independently by Besl and McKay [Besl and McKay 1992] and by Chen and Medioni [Chen and Medioni 1991], is the Iterative Closest Point (ICP) algorithm. Given a source point set $\mathbf{P}$ and a target set $\mathbf{Q}$, ICP minimises:

$$E(\mathbf{R}, \mathbf{t}) = \sum_{i=1}^{N} \|\mathbf{R} \cdot \mathbf{p}_i + \mathbf{t} - \mathbf{q}_i\|^2 \tag{4.2}$$

by alternating between a nearest-neighbour correspondence search and a closed-form transformation update via singular value decomposition, repeating until convergence. Rusinkiewicz and Levoy [Rusinkiewicz and Levoy 2001] provided an influential taxonomy of ICP design choices, demonstrating that the choice of point selection strategy, correspondence metric, and minimisation kernel each independently determine convergence rate and final accuracy. Huang et al. [Huang et al 2021] synthesise the connections between classical optimisation-based approaches and the emerging class of learned registration methods in a comprehensive survey.

Several extensions of ICP improve robustness in the geometrically repetitive, texturally ambiguous structure of vineyard rows. Segal et al. [Segal et al 2009] introduced Generalised ICP, which reformulates the alignment objective as a probabilistic plane-to-plane metric, modelling local surface patches as Gaussian distributions. This formulation is more robust to incorrect correspondences than the standard point-to-point variant and requires less manual tuning of the maximum correspondence distance threshold. He et al. [He et al 2017] proposed GF-ICP, a feature-augmented variant that incorporates point curvature, surface normals, and local density into the correspondence cost. This enriched error metric enables convergence from larger initial errors than pure geometric proximity allows. Choi et al. [Choi et al 2020] extended ICP to coloured point clouds through a probabilistic K-closest matching formulation with a colour-supported objective, adding a photometric constraint that reduces correspondence ambiguity in regions with visually distinct surface texture—a useful property when aligning vine bark or cane surface reconstructions.

Semantic ICP, as developed by Chen et al. [Chen et al 2025], restricts correspondence matching to points sharing the same semantic class label. This prevents the structurally plausible but semantically incorrect matches that arise when multiple anatomical regions are present in a single scan, improving robustness in multi-organ plant reconstructions. Vizzo et al. [Vizzo et al 2023] demonstrated with KISS-ICP that a deliberately lean point-to-point formulation,

augmented by adaptive correspondence thresholding and a robust kernel for outlier rejection, achieves competitive LiDAR odometry accuracy across diverse sensor platforms and environments without requiring per-deployment parameter tuning. Their finding that engineering simplicity can match or exceed more complex variants in practice has informed the design of several subsequent registration systems.

A fundamental limitation common to all ICP variants is sensitivity to the initial pose estimate: convergence to the global optimum of eq. (4.2) is not assured when the initial alignment error exceeds the geometric scale of distinctive scene features. For the more challenging case of global registration from an arbitrary initial pose, Mellado et al. [Mellado et al 2014] introduced Super4PCS, which detects congruent four-point bases across two overlapping point clouds in linear time, enabling alignment in arbitrary initial orientations with as little as 25% overlap. Yang et al. [Yang et al 2016] proposed Go-ICP, which employs a branch-and-bound search over the $SE(3)$ space of rigid transformations, guaranteeing globally optimal ICP solutions by systematically excluding sub-optimal rotation regions at the cost of increased computation.

The most recent direction adapts deep learning to replace hand-crafted descriptors with learned representations. Elbaz et al. [Elbaz et al 2017] trained a deep auto-encoder to encode overlapping local regions of a point cloud into compact feature vectors; matched descriptor pairs initialise a subsequent ICP refinement, enabling robust alignment even under the large initial misalignment typical when registering a close-range scan against a large-scale prior map. Zhang et al. [Zhang et al 2020] provide a systematic review of deep learning-based registration methods, concluding that learned descriptors consistently outperform hand-crafted alternatives on standard benchmarks, in particular under noise and low-overlap conditions. Xu et al. [Xu et al 2025] address the further challenge of cross-source registration—aligning point clouds from heterogeneous sensor types, such as a spinning mechanical LiDAR and a semi-solid-state LiDAR—through a visual-geometric attention module that fuses image and geometric features to establish robust correspondences across point clouds with different density and pattern distributions. A global survey by Yin et al. [Yin et al 2024] identifies the central open challenge in LiDAR-based global localisation as reliably retrieving the correct initial alignment hypothesis within a large-scale map, noting that the dominant operational paradigm—global place retrieval followed by local ICP refinement—remains fragile under seasonal appearance change, a challenge directly relevant to long-term autonomous deployment in vineyards.

4.3 PROPOSED METHODS

The relocalisation pipeline described in this section refines the coarse initial pose estimate delivered by the SLAM system presented in chapter 3 through point cloud registration against a pre-built 3D Gaussian Splatting reference model. Two independent sensing modalities — stereo vision and dense LiDAR — are evaluated within a common architectural framework; both converge on a shared ICP registration back-end, the output of which is used to correct the robot's estimated pose in the ROS 2 transform tree. The approach is outlined in section 4.1.2; the following subsections present each component in detail.

4.3.1 System Overview

The precise-alignment pipeline operates as a one-shot refinement: when the robot halts in front of a target vine, a single localisation correction is computed and applied before any manipulation motion is initiated. The pipeline consists of three cooperating components: a sensor-specific preprocessing stage that transforms raw sensor data into a filtered, downsampled point cloud; a GPU-accelerated rendering stage that synthesises a target point cloud from the 3D Gaussian Splatting (3DGS) model at candidate viewpoints near the initial pose; and a shared ICP registration stage that searches for the optimal rigid alignment between the sensor observation and the rendered reference. The corrected pose is subsequently propagated to the robot's transform tree without modifying the SLAM system's own published transforms.

Two independent sensing modalities are evaluated. The stereo vision pipeline accepts rectified image pairs from one or two ZED stereo cameras, applies the FoundationStereo foundation model [Wen et al 2025] to estimate dense disparity maps, and derives coloured 3D point clouds that are segmented to retain vine trunk structure. The LiDAR pipeline operates on dense point clouds accumulated from a Livox solid-state sensor over a commanded interval, removing the ground plane with Patchwork++ and applying a multi-stage filtering sequence to isolate vine wood. Both pipelines share the same ICP registration module and the same pose correction mechanism; they differ only in their preprocessing steps, which are tailored to the noise characteristics and geometric coverage of the respective sensor.

The initial pose estimate consumed by the pipeline is provided by the ENU-aligned SLAM system described in section 3.3. This pose locates the robot within the global coordinate frame and is used both to select candidate viewpoints for 3DGS rendering and to anchor the final corrected transform in the global frame.

4.3.2 Reference Model: 3D Gaussian Splatting

The localisation target for both sensor modalities is a point cloud synthesised at runtime from the 3DGS model of the target vine. The construction of this model from SfM photogrammetry is described in section 4.1.1; the present subsection addresses how it is used during localisation.

The 3DGS model is loaded once at initialisation and held resident in GPU memory throughout the localisation session. At registration time, the model is rendered at each candidate viewpoint to produce a target point cloud. Rendering is performed using the *taichi-splatting* GPU-accelerated splat rasteriser, which projects the volumetric Gaussian primitives onto a virtual image plane and recovers depth at each pixel; the depth map is then back-projected into a three-dimensional point set in the camera frame of the candidate viewpoint. This per-viewpoint rendering approach avoids pre-computing a single global point cloud, which would require either a fixed viewpoint selection or a coarse global reconstruction that may not capture the local geometry at the resolution required for precise alignment.

After rendering, the point cloud is filtered to retain only points within a specified depth range. Points closer than a near-clip distance are excluded to suppress rendering artefacts on nearby Gaussian primitives, and points beyond a far-clip distance are discarded as their position

estimates are less reliable and they correspond to environmental structure unrelated to the vine under registration.

4.3.3 Stereo Vision Pipeline

The stereo vision pipeline converts raw rectified image pairs from the ZED stereo camera into a filtered point cloud of the vine trunk, suitable for registration against the 3DGS target described in section 4.3.2. The pipeline proceeds through three sequential stages: dense depth estimation from the stereo pair, optional fusion of clouds from a front and rear camera pair, and trunk segmentation to remove non-structural points. The output is passed directly to the shared ICP registration stage described in section 4.3.5.

4.3.3.1 Depth Estimation

Dense depth estimation is performed using FoundationStereo [Wen et al 2025], a stereo foundation model that produces disparity maps with strong zero-shot generalisation across sensor configurations and scene types (see section 4.2.1 for a description of the model and its benchmarked performance). FoundationStereo is applied to rectified image pairs from the ZED Mini stereo camera. Prior to inference, the input images are downscaled by a factor of 0.7 relative to the original sensor resolution.

The disparity map produced by FoundationStereo is converted to metric depth via the standard pinhole relation (see section 4.2.1):

$$Z = \frac{f \cdot B}{d}, \tag{4.3}$$

where Z is the metric depth, f is the focal length of the rectified camera model, B is the stereo baseline, and d is the estimated disparity at each pixel. A far-clip distance of $z_{\text{far}} = 1.0\,\text{m}$ is applied to discard depth estimates beyond the typical operating range between the robot and the target vine. Points with depth below a minimum threshold are also discarded to suppress artefacts close to the camera aperture.

Each pixel for which a valid depth estimate is available is back-projected into a three-dimensional point in the camera frame, using the corresponding RGB values to produce a coloured point cloud. A radius outlier removal filter is subsequently applied to eliminate sparse, isolated points that are not part of any coherent surface.

4.3.3.2 Dual-Camera Point Cloud Fusion

The system supports an optional dual-camera configuration in which a second ZED camera is mounted at the rear of the robot platform, oriented toward the vine from the opposite side. When this configuration is enabled, separate point clouds are produced for the front and rear cameras and merged into a unified observation before trunk segmentation.

The extrinsic relationship between the two cameras is expressed as a rigid transform from the rear camera frame to the front camera frame. Given the transform from the front camera to

a common reference frame, $T_{\text{front}\to\text{ref}}$, and the corresponding transform from the rear camera, $T_{\text{back}\to\text{ref}}$, the back-to-front transform is computed as:

$$T_{\text{back}\to\text{front}} = T^{-1}_{\text{front}\to\text{ref}} \cdot T_{\text{back}\to\text{ref}}. \tag{4.4}$$

The rear camera point cloud is transformed into the front camera frame using $T_{\text{back}\to\text{front}}$, and the two clouds are concatenated. The merged cloud provides greater angular coverage of the vine structure than either camera alone, which is particularly beneficial when vine geometry extends behind the plane of the front camera or when obstruction limits the coverage from a single viewpoint.

4.3.3.3 Trunk Segmentation

A custom trunk segmentation pipeline developed for this work is applied to the merged (or single-camera) point cloud to isolate the vine trunk from background vegetation, noise, and ground returns. The pipeline comprises three sequential stages.

Morphological closing. A voxel-based morphological closing operation fills gaps in the trunk surface caused by specular reflections or missing depth estimates. The input cloud is voxelised at a resolution of 0.02 m, and a spherical closing kernel of radius 0.005 m is applied.

DBSCAN clustering. Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [Ester et al 1996] partitions the closed point cloud into spatially coherent clusters. DBSCAN groups points whose neighbourhood within radius ε contains at least $m_{\min}$ neighbouring points, classifying the remainder as noise. With $\varepsilon = 0.05$ m and $m_{\min} = 200$, the dominant cluster — corresponding to the vine trunk — is retained, and all other clusters and noise points are discarded.

Voxel density filtering. The retained cluster is passed through a voxel density filter that partitions the cloud into voxels of side length 0.02 m and discards any voxel containing fewer than 20 points. This step removes sparsely populated surface regions whose geometric contribution to the ICP objective would otherwise be disproportionate to their informational content.

The combined cloud is capped at 600,000 points by random subsampling prior to the first stage, bounding the worst-case computation time independently of sensor resolution or accumulation duration.

The parameters for the stereo preprocessing pipeline are summarised in table 4.1.

4.3.4 LiDAR Pipeline

The LiDAR pipeline converts raw scans from the Livox solid-state sensor into a filtered point cloud of the vine structure, ready for registration. The pipeline proceeds through three stages: dense scan accumulation to build a spatially rich input cloud, ground plane removal using Patchwork++, and sequential geometric filtering to isolate vine wood. The output is forwarded to the shared ICP registration stage described in section 4.3.5.

Table 4.1: Stereo vision pipeline parameters.

Parameter	Value	Description
Downscale factor	0.7	Input image downscale factor for FoundationStereo inference
z_{far}	1.0 m	Far-clip depth; points beyond this distance are discarded
Closing voxel size	0.02 m	Voxel resolution for the morphological closing operation
Closing radius	0.005 m	Spherical kernel radius for the closing operator
DBSCAN ε	0.05 m	Neighbourhood radius for cluster membership
DBSCAN $m_{\min}$	200	Minimum point count to form a core cluster
Density filter voxel size	0.02 m	Voxel size for the density filtering stage
Density filter min count	20	Minimum points per voxel
Input point cap	600 000	Maximum points admitted to the segmentation pipeline

4.3.4.1 Dense Scan Accumulation

The Livox sensor employs a non-repetitive scanning pattern in which, unlike a conventional spinning LiDAR, the scan lines do not repeat on a fixed cycle. Consequently, a single frame provides sparse and unevenly distributed coverage of the target. A custom dense-mapper ROS 2 node [Macenski et al 2022] developed for this work accumulates multiple consecutive scans over a commanded time interval, concatenating their returns into a single dense point cloud. Over a sufficiently long interval, the non-repetitive pattern achieves high fill-factor coverage of the scene within the sensor's field of view, producing a point cloud with spatial density comparable to that of a multi-return spinning LiDAR.

4.3.4.2 Ground Segmentation

The accumulated point cloud includes returns from the vineyard floor, which are uninformative for vine pose estimation and introduce spurious correspondences during ICP. Ground plane removal is performed using Patchwork++ [Lee et al 2022], an incremental ground segmentation algorithm that models the ground surface as a set of concentric annular patches and applies region-wise plane fitting with outlier rejection. The method is robust to uneven terrain and does not require a flat-ground assumption.

The parameters used in this work are as follows. The sensor height above the ground surface is set to 0.88 m, consistent with the physical mounting position of the Livox sensor on the robot platform. The distance threshold for classifying a point as ground is 0.125 m. The seed threshold for initialising planar patches is 0.5 m, and three plane-fitting iterations are applied per patch. The processing range is bounded to 0.1–3.0 m to discard returns either too close to the sensor aperture or beyond the typical vine operating distance. The Ring-wise Normal-based Removal (RNR) and Ring-wise Vertical Point Filtering (RVPF) modules are enabled to suppress scan-line artefacts characteristic of solid-state LiDAR point clouds.

4.3.4.3 Point Cloud Processing

Following ground removal, the non-ground point cloud is passed through a custom multi-stage filtering pipeline developed for this work. The five sequential stages are described below.

Statistical outlier removal. Each point is evaluated against its k-nearest neighbourhood; points whose mean neighbour distance exceeds the population mean by more than λ standard deviations are classified as outliers and removed. With $k = 16$ and $\lambda = 2.0$, the first pass removes the most isolated returns, which typically correspond to Livox streak artefacts in far-range returns.

Angular filtering. Points in the azimuthal range from $-180°$ to $0°$ are retained; returns outside this half-space are discarded. This step exploits the known mounting orientation of the Livox sensor to exclude returns from the robot body and from the vineyard row that the robot is not currently targeting.

Radial outlier removal. A spherical neighbourhood of radius 0.05 m around each point is examined; points with fewer than 30 neighbours within that radius are discarded. This pass targets point clusters that are spatially consistent at the individual-point level but are too isolated to form part of a coherent surface.

Normal-based filtering. Surface normals are estimated for each point from its 75 nearest neighbours. Points whose estimated normal deviates by more than 100° from the expected surface orientation are discarded.

Normal-based smoothing. A second normal-based pass with 30 nearest neighbours and a 25° angular threshold enforces local normal consistency across the retained surface, smoothing boundary effects and reducing the influence of surface noise on the ICP correspondence stage.

The parameters for the LiDAR preprocessing pipeline are summarised in table 4.2.

4.3.5 ICP Registration Against the 3DGS Model

Both the stereo and LiDAR preprocessing branches deliver a filtered source point cloud to a shared ICP registration module, developed for this work, which aligns the sensor observation against a target point cloud rendered on demand from the 3DGS model. The registration procedure is as follows.

(a) Initial pose. The initial pose estimate, obtained from the SLAM system (section 3.3), is represented as a 4×4 homogeneous transformation matrix in $SE(3)$, encoding the rotation and translation of the robot base in the global frame.

(b) Candidate viewpoint sampling. $N = 5$ candidate viewpoints are sampled in the neighbourhood of the initial pose. Each candidate represents a plausible refined pose from which the 3DGS model is rendered for registration.

(c) Reference cloud rendering. The 3DGS model is rendered at each candidate viewpoint as described in section 4.3.2, yielding N target point clouds.

(d) Downsampling. Both the source cloud and each target cloud are independently downsampled using a voxel grid filter with voxel side length $v = 0.0075$ m. This step reduces

Table 4.2: LiDAR pipeline parameters.

Parameter	**Value**	**Description**
Patchwork++ ground segmentation		
Sensor height	0.88 m	Mounting height of the Livox sensor above the ground
Distance threshold	0.125 m	Maximum deviation from the fit plane to be classified as ground
Seed threshold	0.5 m	Height threshold for initial ground seed selection
Fitting iterations	3	Number of plane-fitting iterations per annular patch
Processing range	0.1–3.0 m	Minimum and maximum range included in ground estimation
RNR	Enabled	Ring-wise Normal-based Removal
RVPF	Enabled	Ring-wise Vertical Point Filtering
Point cloud filtering		
Statistical outlier k	16	Neighbourhood size for statistical outlier removal
Statistical outlier λ	2.0	Standard deviation multiplier for the outlier threshold
Angular filter range	-180° to 0°	Retained azimuthal range
Radial outlier radius	0.05 m	Neighbourhood radius for radial outlier removal
Radial outlier min neighbours	30	Minimum neighbours within the search radius
Normal filter k	75	Neighbourhood size for normal estimation (filtering stage)
Normal filter threshold	100°	Maximum normal deviation for point retention
Normal smooth k	30	Neighbourhood size for normal estimation (smoothing stage)
Normal smooth threshold	25°	Maximum normal deviation for smoothing-stage retention

the computational cost of each ICP iteration while retaining sufficient geometric detail for convergence.

(e) ICP. ICP is run between the downsampled source and each downsampled target using the Open3D [Zhou et al 2018] tensor-based point-to-point implementation. The ICP objective (eq. (4.2)) is minimised iteratively until either the maximum iteration count is reached or the relative change in fitness and RMSE falls below the convergence thresholds stated in table 4.3.

(f) Candidate selection. The registration result with the highest fitness score — defined as the ratio of accepted correspondences to total source points — and the lowest point-to-point RMSE is selected as the final alignment across all N candidates.

The output of the registration stage is a result object containing the estimated transformation matrix, the fitness score, the point-to-point RMSE, and the accepted correspondence set.

LiDAR-Specific Robust Kernel

When the source cloud originates from the LiDAR pipeline, a Cauchy robust loss function is applied to the ICP objective to reduce the influence of outlier correspondences. The Cauchy loss is defined as:

$$\rho(r) = \frac{k^2}{2} \ln\left(1 + \left(\frac{r}{k}\right)^2\right), \tag{4.5}$$

where r is the residual distance between a source point and its corresponding target point, and $k = 0.15$ is the scale parameter that controls the transition from approximately quadratic behaviour near the origin to logarithmic growth in the outlier region. The standard point-to-point objective without a robust kernel is used for the stereo modality, where the trunk segmentation pipeline substantially reduces the outlier fraction prior to registration.

The parameters for the ICP registration stage are summarised in table 4.3.

4.3.6 Coordinate Frames and Pose Correction

The ICP registration stage returns a rigid transformation expressed in the sensor frame — that is, the frame of the camera or LiDAR sensor from which the source cloud was acquired. Before this result can be applied to the robot's localisation, it must be expressed in the base link frame that serves as the canonical reference for all downstream planning and control components, and it must be introduced into the ROS 2 transform tree [Macenski et al 2022] in a manner that does not disturb the SLAM system's own transform publications.

Coordinate Frame Hierarchy

The coordinate frame hierarchy used throughout this system is:

$$\texttt{local_enu} \rightarrow \texttt{map} \rightarrow \texttt{odom} \rightarrow \texttt{base_link} \rightarrow \texttt{sensor_frame},$$

Table 4.3: ICP registration parameters.

Parameter	**Value**	**Description**
Candidate viewpoints (N)	5	Number of candidate viewpoints sampled around the initial pose
Voxel grid size (v)	0.0075 m	Downsampling resolution applied to source and target clouds
ICP backend	Open3D `t`	Tensor-based point-to-point ICP implementation
Max iterations	*TODO*	Maximum number of ICP iterations per candidate
Max correspondence distance	*TODO*	Maximum point-pair distance for correspondence acceptance
Relative fitness	*TODO*	Convergence criterion: minimum relative change in fitness
Relative RMSE	*TODO*	Convergence criterion: minimum relative change in RMSE
Robust kernel (LiDAR only)	Cauchy, $k = 0.15$	Applied to LiDAR source clouds; not used for stereo

where `local_enu` is the ENU-anchored global frame, `map` and `odom` are the frames published by the SLAM system, `base_link` is the robot body frame, and `sensor_frame` is the frame of whichever sensor produced the source cloud. The static transforms between `base_link` and the sensor frames are fixed by the mechanical configuration of the platform and are loaded at launch from the robot description.

TF Correction Mechanism

A custom TF correction mechanism developed for this work inserts a correction frame into the transform tree without modifying any of the transforms published by the SLAM system. The ICP result provides the correction in the sensor frame, $T_{\text{correction,sensor}}$. To express this correction at the level of the base link, the following composition is applied:

$$T_{\text{correction,base}} = T^{-1}_{\text{base}\to\text{sensor}} \cdot T_{\text{correction,sensor}} \cdot T_{\text{base}\to\text{sensor}}, \tag{4.6}$$

where $T_{\text{base}\to\text{sensor}}$ is the fixed extrinsic transform from the base link to the sensor frame. The resulting base-frame correction is then inserted into the appropriate location in the transform chain and published as a static transform.

Because the correction is published as a static transform, it does not require a continuous update loop and is non-disruptive to the SLAM publishers. The correction mechanism supports both automatic triggering — applied immediately upon successful ICP convergence — and manual triggering, in which the operator confirms before the correction is broadcast to the transform tree. The appropriate mode is selected at launch via a configuration parameter.

4.3.7 Pipeline Execution Model

Each localisation request is encapsulated as an ordered sequence of command objects following the command design pattern. Each processing stage — depth estimation, point cloud preprocessing, 3DGS rendering, ICP registration, and pose correction — is implemented as an `AsyncCommand` with a well-defined input contract and output contract. Commands are submitted to an asynchronous task queue, which executes them sequentially within a dedicated worker thread pool. This design decouples the processing pipeline from the ROS 2 executor event loop: computationally intensive operations such as FoundationStereo inference and Open3D ICP run in non-blocking threads, avoiding latency spikes in the ROS 2 message-handling callbacks.

GPU-resident operations — specifically FoundationStereo inference and 3DGS rendering — are governed by a semaphore-based concurrency guard that prevents simultaneous kernel execution from exceeding available GPU memory. When a new localisation request arrives while a prior job is still processing, the new job is queued independently and executes after the in-progress job completes, ensuring that each sensor observation is processed in full without resource contention. This execution model supports both the stereo and LiDAR pipelines without modification, as the modality-specific processing difference is captured entirely within the command sequence submitted for each observation type.

The performance of both sensing modalities across a set of vineyard test cases is evaluated in section 4.4.

4.4 RESULTS

illustrate results

4.5 DISCUSSION

discuss results

4.6 SUMMARY

Summarise findings from this chapter

Chapter 5

CONCLUSION

Highlight key findings. Link back to Introduction

5.1 FUTURE WORK

What could be extended upon in future work?

Chapter 6

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SUPPLEMENTARY FIGURES